

Effects at a Quantile of the Outcome

A two-step percentile-weights estimator, with analytical and survey-design inference

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Abstract

Percentile weights regression (PWR) measures how the effect of the covariates on an outcome varies along the distribution of that outcome. Its device is a Gaussian kernel on the rank of the outcome, centred at a target percentile τ , which isolates the units that sit at that quantile. This paper gives the method in its two-step form. The effect at a quantile is an average — the mean effect of the units there — so PWR estimates it as one: a model of the heterogeneity of the effect gives every unit its own effect, and the percentile weights average those effects over the group at τ .

Two heterogeneity models are available: interactions with observables, and a conditional quantile regression. On observables the estimator has a closed form and an exact decomposition of the profile across quantiles into the composition of the group and the heterogeneity of the effect, and the same weights deliver the effect at the quantile of the outcome *before* the contribution of a chosen covariate. The paper derives the influence-function variance, its Taylor linearisation under a complex survey design, and the mean-squared-error-optimal bandwidth of the averaging step, which the uniform rank puts in closed form. The percentile weights are shown to be the density ratio $f(q_\tau|x)/f(q_\tau)$, obtained without estimating a density, so the estimator converges to the unconditional quantile partial effect. As the group at a quantile is defined by the outcome, the profile across quantiles mixes composition with heterogeneity; the effect for a given household separates them, with a simultaneous band and a test of constancy.

Earlier statements of PWR (Araar, 2016, 2023, 2026) applied the percentile weights to a regression of the outcome on the covariates. Proposition 1 shows why they belong on the averaging instead: inside a window on the outcome the outcome barely varies, so the weighted slope is of the order of the squared bandwidth and carries the slope of $\mathbb{E}[x|y]$, not the effect. On the Burkina Faso 1998 survey it is one to two orders of magnitude below the two-step effects; in the Monte Carlo the two-step estimator on observables is at the truth where its heterogeneity model holds, and several times more precise than RIF regression.

JEL codes: C14, C21, C51.

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1 Introduction

Whether the effect of a covariate on an outcome is the same at the bottom and at the top of the distribution of that outcome is the question that quantile methods were built to answer. Conditional quantile regression (Koenker and Bassett, 1978) answers it for the conditional distribution given the covariates; unconditional quantile regression (Firpo et al., 2009) answers it for the marginal distribution, by regressing the recentered influence function (RIF) of the quantile on the covariates. The percentile weights regression (Araar, 2016, 2023) was proposed as a third route with a direct intuition: give each household a kernel weight according to the distance between its rank in the outcome and the target percentile τ , and run the regression with those weights, so that the estimation “focuses on” the households at τ .

The intuition selects the right households. It does not measure the effect among them. A window on the outcome is, in the limit, a level set of the outcome: inside it y is constant, and a regression of a constant on x has slope zero. With a finite bandwidth the outcome varies a little inside the window, and the weighted slope is a small number proportional to the squared bandwidth and to the slope of the reverse regression $\mathbb{E}[x | y]$ at the quantile, a number that says which households are slightly above or below the quantile, not what a change in x does to y . This is the classical consequence of selecting on the dependent variable (Hausman and Wise, 1977; Heckman, 1979), and under normality Goldberger (1981) gives the exact attenuation factor. On the Burkina Faso survey used throughout this paper, the one-step estimates are one to two orders of magnitude below the effects. The examples of the 2016 note and of the 2023 working paper did not reveal this because in all of them the outcome is a deterministic monotone function of a single regressor: the households at the τ -quantile of y are then the households at the τ -quantile of x , the window is a segment of a curve rather than a level set, and the weighted slope is the total derivative along the curve.

This paper states the diagnosis, with the derivation and the simulations that establish it, and replaces the estimator. The replacement keeps the percentile weights, which are the right weights, and moves them from the regression to the averaging. A first step estimates the effect of x for each household from the whole sample, where y varies, with a heterogeneity model: interactions of the regressors with observables, or a conditional quantile regression that lets the effect vary with the household’s conditional rank, which is the route of Alejo et al. (2025). A second step averages those household-level effects with the percentile weights at τ . The weighted average converges to the expected effect of the households at the τ -quantile of y , which is also the derivative of the population quantile with respect to a marginal shift of x : the unconditional quantile partial effect (UQPE) of Firpo et al. (2009). The percentile weights turn out to be their weights $f(q_\tau|x)/f(q_\tau)$, estimated by the rank of y instead of a conditional density. On observables the estimator has a closed form, an exact decomposition of the profile of the effect across quantiles into the composition of the group at each quantile and the heterogeneity of the effect, an influence-function variance that carries the sampling variability of the ranks, the Taylor linearisation of that variance under stratified cluster sampling, and an equivalent stacked-GMM representation. The same weights on the rank of the outcome purged of the contribution of a covariate give the effect at the *initial* quantile, the object of the programme evaluation literature on the distribution of impacts (Heckman et al., 1997). The ranking on a covariate other than the outcome (`rankvar`) is a varying-coefficient regression along that covariate; it selects on a regressor, not on the outcome, and is unaffected.

The profile of the effect across quantiles needs a caution of its own. The group at τ is defined by the outcome, so it changes with τ , and a profile that moves may say only that the households found at the top are not those found at the bottom. Section 7 separates the composition of the group from the heterogeneity of the effect by following a given household along the distribution: under the observables model its effect does not move, by construction, and the whole profile is composition; under the conditional-rank model it moves with the household's conditional rank, and its estimator is the matched coefficient of [Alejo et al. \(2025\)](#) evaluated for that household. A correction of the profile by a linear projection, the obvious alternative, is shown not to be consistent. The same section gives the profile a simultaneous confidence band and a test that the effect is constant across quantiles.

2 Three objects

Let $(y_i, \mathbf{x}_i, \mathbf{z}_i, f_i)$, $i = 1, \dots, n$, be a sample of households with outcome y_i , regressors $\mathbf{x}_i \in \mathbb{R}^k$, further observables $\mathbf{z}_i \in \mathbb{R}^m$ (which may include regressors) and sampling weights f_i , normalised to mean one. Write the outcome as

$$y_i = m(\mathbf{x}_i, u_i), \quad b_{ij} = \frac{\partial m(\mathbf{x}_i, u_i)}{\partial x_j}, \quad (1)$$

where u_i collects everything that is not in $\mathbf{x}_i$ and b_{ij} is the effect of x_j for household i : a derivative for a continuous regressor, a difference $m(\dots, 1, \dots, u_i) - m(\dots, 0, \dots, u_i)$ for a binary one. The random-coefficient form $y_i = a_i + \mathbf{x}_i' \mathbf{b}_i$ is the case where m is linear in $\mathbf{x}$ for each household. Let F be the distribution of y , $q_\tau = F^{-1}(\tau)$ its τ -quantile, f its density, and $p_i = F(y_i)$ the rank of household i .

Definition 1 (Effect at a quantile of the outcome). $\theta_j(\tau) = \mathbb{E}[b_{ij} \mid y_i = q_\tau]$, the expected effect of x_j among the households at the τ -quantile of the outcome.

The object has a second reading. If x_j rises by t for every household, the outcome of household i becomes $y_i + t b_{ij} + o(t)$, and the τ -quantile of the new distribution moves at the rate

$$\left. \frac{d}{dt} Q_\tau(y + t b_j) \right|_{t=0} = \mathbb{E}[b_{ij} \mid y_i = q_\tau] = \theta_j(\tau), \quad (2)$$

under continuity of f at q_τ and of $\mathbb{E}[b_j \mid y]$ at q_τ (Appendix A). The right-hand side is the unconditional quantile partial effect of [Firpo et al. \(2009\)](#), who write it as

$$\text{UQPE}_j(\tau) = \mathbb{E}[\omega_\tau(\mathbf{x}_i) \mathbb{E}[b_{ij} \mid \mathbf{x}_i, y_i = q_\tau]], \quad \omega_\tau(\mathbf{x}) = \frac{f(q_\tau \mid \mathbf{x})}{f(q_\tau)}, \quad (3)$$

by conditioning on $\mathbf{x}$: the effect at the quantile is the average over households of their type-specific effect at the quantile, weighted by how likely each type is to be found at q_τ . Equations (2) and (3) are the same number. RIF regression estimates it by modelling $\Pr[y \leq q_\tau \mid \mathbf{x}]$; the two-step estimator of Section 4 estimates it by modelling b_{ij} and estimating ω_τ with the ranks.

Definition 2 (Effect at the initial quantile). For a set J of regressors and reference values x_j^{ref} , let $y_{0i} = y_i - \sum_{j \in J} b_{ij} (x_{ij} - x_j^{\text{ref}})$ be the outcome of household i before the contribution of the regressors in J , and $q_{0\tau}$ the τ -quantile of y_0 . Then $\theta_{0j}(\tau) = \mathbb{E}[b_{ij} \mid y_{0i} = q_{0\tau}]$.

This is the effect for the households that were at the τ -quantile before the regressors in J acted, the object of “who gains among the initially poor” (Heckman et al., 1997). For a programme indicator with $x^{\text{ref}} = 0$, y_0 is the outcome without the programme. It requires a model of the household-level effect, the same model as the two-step estimator, and a reference value; the households are re-ranked on y_0 , so the group at τ is not the group at τ of y .

Definition 3 (Profile along a covariate). For an observed z that is not the outcome, $\theta_{zj}(\tau) = \mathbb{E}[b_{ij} \mid z_i = q_\tau^z]$, the expected effect of x_j among the households at the τ -quantile of z .

The profile along household size, age of the head, assets, or a predicted welfare index $\mathbf{x}'\hat{\beta}$ is a varying-coefficient regression in z . It selects households on a covariate, and a regression of y on $\mathbf{x}$ among households with $z \approx q_\tau^z$ measures the effect there, because y varies inside that group. This is the object of `rankvar(z)` in `gepwreg` from the start, and nothing in this paper changes it.

Two remarks on the first object. It is defined on the observed outcome, so the group at τ is the group observed there, with the covariates it has: a household is at the τ -quantile in part because of its x . That is the question $\theta_j(\tau)$ answers. What it cannot be obtained from is a regression run inside the group, and Section 3 is about why.

2.1 Names for the estimators

Six estimators appear in what follows; Table 1 is the key. The name PWR alone, without an index, is what Araar (2016, 2023) call the estimator, and it is PWR-1 here.

Table 1: The estimators, their names in this paper, and the syntax that produces them

Name	Object	<code>gepwreg</code>	Where
PWR-1	none: a descriptive slope	<code>rankdep</code>	§3
PWR-2(z)	$\theta_j(\tau)$, Def. 1	<code>het(varlist)</code>	§4.2
PWR-2(qr)	$\theta_j(\tau)$, Def. 1	<code>het(qr)</code>	§4.3
PWR-2(y_0)	$\theta_{0j}(\tau)$, Def. 2	<code>initial(varlist)</code>	§4.8
PWR-cov(z)	$\theta_{zj}(\tau)$, Def. 3	<code>rankvar(z)</code>	§4.9
PWR-ref	$\theta_j^\circ(\tau)$, Def. 4	<code>ref()</code>	§7.2
RIF-OLS	$\theta_j(\tau)$, by a different route	<code>rifhdreg</code> (Rios-Avila, 2020, 2018)	§4.5

PWR-1 is the estimator of Araar (2016, 2023) and of `gepwreg` up to version 1.3; Section 3 shows that it does not estimate $\theta_j(\tau)$, and it is retained as a descriptive statistic. PWR-2(z) and PWR-2(qr) differ only in the model that gives each household its effect, and target the same object. PWR-2(qr) is the estimator of Alejo et al. (2025); see Section 4.3. PWR-ref is the effect for a given household, under either model; see Section 7.

Implementation. Every estimator in Table 1 is implemented in `gepwreg` 1.5.0 (Araar, 2026b), with the analytical and survey-design standard errors of Section 6 and the bandwidth of Section 4.10. Its help file documents the syntax, the options and the stored results, and is the reference that stays current with the package; the command itself is the subject of a separate paper. The estimates reported below were produced with version 1.4, which version 1.5.0 reproduces, except those of Sections 7 and 9.6, which use options new

in version 1.5.0 (the effect for a given household, the simultaneous band and the test of constancy).

3 What the outcome-ranked regression measures

3.1 The outcome-ranked estimator

Let $\hat{F}_n(y) = \sum_i f_i \mathbf{1}[y_i \leq y] / \sum_i f_i$ be the weighted empirical distribution of the outcome, $\hat{p}_i = \hat{F}_n(y_i)$ the rank of household i , and

$$w_i(\tau) = K\left(\frac{\hat{p}_i - \tau}{h}\right), \quad K(u) = \exp(-u^2/4), \quad (4)$$

the percentile weight of household i at the target τ , a Gaussian kernel of variance $\sigma_K^2 = 2$ in units of h on the rank scale. In the 2016 and 2023 notes the bandwidth is Silverman's rule (Silverman, 1986) on the ranks, $h = 0.9 \min\{\text{sd}(\hat{p}), \text{IQR}(\hat{p})/1.34\} n^{-1/5} \approx 0.26 n^{-1/5}$, which `gepwreg` keeps as the option `silverman`; version 1.4 uses the plug-in of Section 4.10 for every estimator, this one included. The kernel effective sample size $n_{\text{eff}} = (\sum_i f_i w_i)^2 / \sum_i f_i^2 w_i^2$ is the number of households that count at τ . The estimator of `gepwreg` (and of the 2016 and 2023 notes) is the weighted least squares regression

$$\hat{b}_w(\tau) = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\mathbf{y}, \quad \mathbf{W} = \text{diag}\{f_1 w_1(\tau), \dots, f_n w_n(\tau)\}, \quad (5)$$

which we call the one-step estimator. In `gepwreg` it is available as `rankdep` and labelled as a descriptive slope.

3.2 The population limit

Take one regressor for clarity; the vector case is in Appendix A. Let $m(y) = \mathbb{E}[x | y]$ be the reverse regression and $v(y) = \text{Var}(x | y)$ the conditional variance of the regressor given the outcome. As $n \rightarrow \infty$ with h fixed, $\hat{b}_w(\tau)$ converges to $b_w(\tau, h) = \text{Cov}_w(x, y) / \text{Var}_w(x)$, where Cov_w and Var_w are moments under the weighting $K((p - \tau)/h)$ on the rank $p = F(y)$.

Proposition 1 (Limit of the one-step estimator). *Suppose f is continuous and positive at q_τ , m is differentiable at q_τ and v is continuous at q_τ with $v(q_\tau) > 0$. Then, as $h \rightarrow 0$,*

$$b_w(\tau, h) = \sigma_K^2 h^2 \frac{m'(q_\tau)}{f(q_\tau)^2 v(q_\tau)} + O(h^4). \quad (6)$$

Moreover, for every h , $|b_w(\tau, h)| \leq \text{sd}_w(y) / \text{sd}_w(x)$, and $\text{sd}_w(y) = O(h)$.

Three things follow. The limit goes to zero with the bandwidth: the better the window isolates the group at τ , the smaller the slope, and no bias–variance trade-off has this shape. The limit contains the reverse regression $m(y) = \mathbb{E}[x | y]$ and the conditional variance $v(y)$, features of the joint distribution of (x, y) , and does not contain the effect b_i : two data-generating processes with the same joint distribution of (x, y) and different effects give the same b_w . And its sign is the sign of $m'(q_\tau)$, so what the one-step estimator reports is whether the households slightly above the quantile have more x than the households slightly below it. That is a description of the neighbourhood of the quantile; it is not the effect of x .

The mechanism is elementary. Conditional on the rank p , $y = F^{-1}(p)$ is fixed, so the weighted covariance of x and y is the weighted covariance of $m(p)$ and y , both of which move only through p ; inside a window of width h on the rank they move by $O(h)$, and their covariance is $O(h^2)$. The weighted variance of x does not shrink, because x varies freely given y : it converges to $v(q_\tau)$. The ratio is $O(h^2)$. This is selection on the dependent variable (Hausman and Wise, 1977; Heckman, 1979) taken to its limit.

3.3 The Goldberger identity

When the reverse regression is linear and homoskedastic, the attenuation has an exact expression that does not depend on the shape of the weights.

Proposition 2 (Goldberger, 1981). *Suppose $\mathbb{E}[x | y] = a + \gamma y$ and $\text{Var}(x | y) = v$ for all y , as under bivariate normality. Let $\beta = \text{Cov}(x, y)/\text{Var}(x)$ be the population OLS slope and b_w the slope under any weighting that depends on y alone. Then*

$$\frac{b_w}{\beta} = \frac{\text{Var}_w(y)/\text{Var}(y)}{\text{Var}_w(x)/\text{Var}(x)}. \quad (7)$$

Selection on y retains little of $\text{Var}(y)$ and most of $\text{Var}(x)$, hence the attenuation. Table 2 verifies the identity on the DGP $y = 5 + 0.5x + e$ with $x, e \sim \mathcal{N}(0, 1)$ and $n = 100,000$, for bandwidths from 0.02 to 3: the two sides agree to the second decimal at every h (the largest gap, 0.007, is the sampling error of one sample). At the Silverman bandwidth on this sample, $h = 0.026$, the retained share of $\text{Var}(y)$ is below 0.01; the one-step estimator returns about one percent of the effect. As h grows the weights flatten, both shares go to one and the one-step estimator becomes OLS, which is the limit visible in the last rows.

Table 2: The Goldberger identity, DGP A ($y = 5 + 0.5x + e$, $n = 100,000$)

h	n_{eff}	$\text{Var}_w(y)/\text{Var}(y)$	$\text{Var}_w(x)/\text{Var}(x)$	Ratio	b_w/β
0.02	10,027	0.0049	0.7883	0.0063	0.0070
0.05	25,066	0.0320	0.8012	0.0399	0.0364
0.1	50,092	0.1479	0.8282	0.1786	0.1716
0.2	86,474	0.5567	0.9111	0.6110	0.6063
0.3	96,406	0.7715	0.9542	0.8085	0.8049
0.5	99,474	0.9116	0.9823	0.9281	0.9250
1	99,966	0.9773	0.9954	0.9817	0.9789
3	100,000	0.9975	0.9995	0.9980	0.9952

Ratio: right-hand side of (7). b_w/β : one-step slope at $\tau = 0.5$ with bandwidth h divided by the OLS slope. Weighted variances use the percentile weights (4). File `p1onestep_attenuation.do`.

3.4 The exception: an outcome that is a function of one regressor

Proposition 1 requires $v(q_\tau) > 0$. When $y = g(x)$ with g strictly monotone and differentiable, $v \equiv 0$, the rank of y is the rank of x , and the window at τ on y is a window at τ on x . Inside it y varies with x along the curve, and the weighted slope converges to $g'(q_\tau^x)$: the derivative of y with respect to x at the τ -quantile of x , which is here the total derivative of y along the curve. With $y = g(x) + \varepsilon$, the estimator moves continuously from g' to the attenuated limit (6) as $\text{sd}(x | y)$ grows from zero.

The condition fails, and the estimator agrees with the effect, exactly when the outcome is close to a function of a single regressor. That is a strong requirement on the data: it needs the residual dispersion of x given y to be a small fraction of the dispersion of x , which no household survey delivers — on the Burkina Faso data of Section 9 it is nearly one. Where it does hold, the estimator returns the total derivative of y along the curve, which is not the effect of the covariate holding the rest fixed, and the two coincide only when the coefficient does not vary with the regressor itself.

3.5 Why no correction repairs it

Proposition 1 makes the failure structural rather than accidental, so the repairs that come to mind are worth ruling out explicitly. The limit is a product: a shrinking factor times a quantity that is not the effect. A repair must therefore either undo the shrinking, or change the sample the regression is run on. Both routes are closed, and for different reasons.

Dividing by the attenuation factor returns OLS, not the effect at τ . The Goldberger ratio (7) is estimable, so divide the one-step slope by it. Under the conditions of Proposition 2 the corrected slope does converge, and it is near 0.5 on DGP A (between 0.45 and 0.64, with the noise of a ratio of two small numbers). But those conditions — a linear reverse regression with constant $\text{Var}(x|y)$ — hold under a *common* effect, and fail as soon as the effect varies: with a scale effect (B and C) the corrected slope is -1.7 for -0.11 and 1.1 for 0.20 ; with an effect that rises with y (D) it is inverted, 1.6 at the bottom for 1.47 and 1.3 at the top for 1.59 . And where it does work, what it recovers is the common slope, which the full-sample regression already gives with a smaller variance. A correction built on a common effect cannot reveal an effect that varies across quantiles.

Reweighting the covariates does not undo selection on the outcome. Weighting the selected sample by $w_i/\mathbb{E}[w_i|\mathbf{x}_i]$ restores the covariate distribution of the population, which is the standard cure when the selection is on $\mathbf{x}$. Here it is not: inside the window the residual is tied to x mechanically — in the hard band 0.35 – 0.45 of y on DGP A the correlation between x and the error is -0.98 — and restoring the marginal distribution of x does not make y vary among households that were selected for having the same y . On DGP A the double-weighted slope is indistinguishable from the one-step slope; on B, C and D it moves, by up to a factor of ten, and stays far from the effect.

The truncated likelihood identifies the effect, at a rate that makes it useless. On the band $0.35 \leq F(y) \leq 0.45$ of DGP A, with 10,000 households, the truncated-normal maximum likelihood of Hausman and Wise (1977) is correctly specified and consistent, and it returns 0.50 with a standard error of 1.97 . The point is not that the band fails to identify β — the curvature of $\log f_Y$ inside the band carries $\text{Var}(y)$, which is exactly what that likelihood exploits, so β is identified. It is that a random subsample of the same size would give a standard error of 0.010 : the band holds about one part in forty thousand of the information.

What does work is to select on a covariate instead. Restricting the same DGP to the band 0.25 – 0.75 of x leaves OLS at 0.47 (s.e. 0.012), and to the band 0.35 – 0.45 at 0.67 (s.e. 0.13): selecting on a regressor costs precision and no consistency, because y still varies inside the group. This is why `rankvar(z)` measures what it claims, and it is the principle

the two-step estimator generalises — measure the effect where y varies, then carry it to the group at τ with the percentile weights.

3.6 Monte Carlo

Table 3 reports the one-step estimator with the MSE-optimal bandwidth on four DGPs with $n = 100,000$, against the mean effect of the households whose rank in y is within ± 0.05 of τ . With a constant effect (A) the estimator is between 0.002 and 0.04 for 0.5; with a scale effect (B, C), where the effect at a quantile runs from -0.1 to 1.2 , it is between 0.001 and 0.1 with the right sign; with an effect that rises with the unobserved U (D), from 0.02 to 0.63 for 1.45–1.59. The MSE-optimal bandwidth moves between 0.011 and 0.058, and the estimate moves with it, as (6) says: sweeping h from 0.005 to 2, the one-step estimator follows h^2 from less than a thousandth of the effect up to the OLS coefficient. There is no bandwidth at which it is the effect at the quantile, because at small h it is the attenuated slope and at large h it is OLS.

Table 3: The one-step estimator against the effect at the quantile ($n = 100,000$, MSE-optimal bandwidth)

DGP	τ	Truth	One-step	s.e. (IF)	h	n_{eff}	OLS
A: $y = 5 + 0.5x + e$	0.1	0.500	0.0382	0.0019	0.027	13,505	0.499
	0.25	0.500	0.0029	0.0006	0.013	6,733	0.499
	0.5	0.500	0.0247	0.0011	0.058	28,967	0.499
	0.75	0.500	0.0020	0.0006	0.013	6,699	0.499
	0.9	0.500	0.0376	0.0019	0.027	13,394	0.499
B: $y = 5 + 0.5x + e^{0.5x}e$	0.1	-0.109	-0.0140	0.0010	0.017	8,590	0.493
	0.25	0.203	0.0020	0.0005	0.014	6,805	0.493
	0.5	0.464	0.0047	0.0007	0.020	9,990	0.493
	0.75	0.785	0.0053	0.0010	0.011	5,453	0.493
	0.9	1.196	0.1017	0.0039	0.019	9,404	0.493
C: $y = 5 + 0.5x + e^{-0.5x}e$	0.1	1.190	0.1052	0.0041	0.019	9,533	0.504
	0.25	0.782	0.0047	0.0010	0.011	5,444	0.504
	0.5	0.468	0.0058	0.0007	0.021	10,610	0.504
	0.75	0.206	0.0007	0.0005	0.014	6,781	0.504
	0.9	-0.100	-0.0150	0.0010	0.018	8,906	0.504
D: $y = 5 + (1 + U)x + U + e$	0.1	1.468	0.3023	0.0057	0.029	14,464	1.496
	0.25	1.450	0.0183	0.0016	0.013	6,664	1.496
	0.5	1.462	0.1188	0.0026	0.049	24,669	1.496
	0.75	1.510	0.0161	0.0016	0.013	6,388	1.496
	0.9	1.592	0.6307	0.0084	0.052	22,329	1.496

$x, e \sim \mathcal{N}(0, 1)$, $U \sim U(0, 1)$. Effects: A, 0.5; B, $0.5 + 0.5e^{0.5x}e$; C, $0.5 - 0.5e^{-0.5x}e$; D, $1 + U$. Truth: mean effect of the households whose rank in y is within ± 0.05 of τ . s.e. (IF): the influence-function standard error of the earlier paper, which is correct for the quantity the one-step estimates. File `p1.onestep.attenuation.do`.

4 A two-step estimator of the effect at a quantile

4.1 The estimator and what the percentile weights are

The one-step estimator fails because it asks the group at τ to reveal the effect from its own variation in y , which it does not have. The two-step estimator measures the effect where y varies, for each household, and brings it to the group with the percentile weights:

$$\hat{\theta}_j(\tau) = \frac{\sum_i f_i w_i(\tau) \hat{b}_{ij}}{\sum_i f_i w_i(\tau)}, \quad (8)$$

where $\hat{b}_{ij}$ is an estimate of the effect of x_j for household i from a heterogeneity model fitted on the whole sample (Sections 4.2 and 4.3), and $w_i(\tau)$ are the percentile weights (4). The weights do the selection; the model does the measurement.

Proposition 3 (The percentile weights). *Let b_i be any household-level quantity with $g(p) = \mathbb{E}[b_i | p_i = p]$ twice differentiable at τ , and let $\omega_\tau(\mathbf{x}) = f(q_\tau | \mathbf{x}) / f(q_\tau)$. As $n \rightarrow \infty$ and then $h \rightarrow 0$,*

$$\frac{\sum_i f_i w_i(\tau) b_i}{\sum_i f_i w_i(\tau)} \longrightarrow g(\tau) + \frac{\sigma_K^2 h^2}{2} g''(\tau) + O(h^4), \quad g(\tau) = \mathbb{E}[b_i | y_i = q_\tau], \quad (9)$$

$$\frac{\mathbb{E}[w_i(\tau) | \mathbf{x}_i]}{\mathbb{E}[w_i(\tau)]} \longrightarrow \omega_\tau(\mathbf{x}_i). \quad (10)$$

The first line says that a percentile-weighted mean of any household-level quantity converges to its expectation in the group at τ , with a smoothing term proportional to the curvature of its profile across ranks: at the Silverman bandwidth of the Burkina Faso sample ($h = 0.043$), $\sigma_K^2 h^2 / 2 = 0.0018$, so a profile would have to bend sharply for the term to matter. The second line says that the expected weight of a household of type $\mathbf{x}$, relative to the average weight, is the density ratio of [Firpo et al. \(2009\)](#): the percentile weights are their weights, obtained from the rank of y without an estimate of a conditional density. Applied to $b_i = b_{ij}$, (9) is Definition 1; by (3) it is the UQPE. The two-step estimator therefore has the same estimand as RIF regression, with a different auxiliary model (Section 4.5).

The bandwidth defines the width of the group. It is not chosen to trade the bias of a slope against its variance, because there is no slope here; but (8) is a smoother, and the trade-off it does face — the curvature of the profile $\theta_j(\tau)$ against the dispersion of the effects within the group — is derived in Section 4.10. The plug-in solution of that trade-off is the default; `silverman` restores the rank-scale Silverman rule and `band()` or `cband()` widen or narrow the group by hand.

4.2 Heterogeneity on observables: `het(z)`

Let $\mathbf{z}_i$ be the observables through which the effect is allowed to vary, which may include regressors. The first step is the weighted least squares regression, on the whole sample,

$$y_i = \alpha + \mathbf{x}_i' \boldsymbol{\gamma} + \mathbf{z}_i' \mathbf{c} + \sum_{j=1}^k \sum_{l=1}^m x_{ij} z_{il} d_{jl} + e_i, \quad \mathbb{E}[e_i | \mathbf{x}_i, \mathbf{z}_i] = 0, \quad (11)$$

whose derivative with respect to x_j is the household effect

$$\hat{b}_{ij} = \hat{\gamma}_j + \sum_{l=1}^m \hat{d}_{jl} z_{il} + \sum_{l: z_l \equiv x_j} \left(\hat{c}_l + \sum_{j'=1}^k \hat{d}_{j'l} x_{ij'} \right), \quad (12)$$

where the last sum collects, when x_j is itself one of the heterogeneity variables, the derivative of the terms in which it enters as z_l (the quadratic and the cross terms). For a binary x_j the effect is the difference of (1), not a derivative: since $x_{ij}^2 = x_{ij}$, the quadratic term contributes $\hat{d}_{jl}$ where its derivative would give $2\hat{d}_{jl}x_{ij}$. Substituting (12) in (8) gives the closed form

$$\hat{\theta}_j(\tau) = \hat{\gamma}_j + \sum_{l=1}^m \hat{d}_{jl} \bar{z}_l(\tau) + \sum_{l: z_l \equiv x_j} \left(\hat{c}_l + \sum_{j'} \hat{d}_{j'l} \bar{x}_{j'}(\tau) \right), \quad \bar{z}_l(\tau) = \frac{\sum_i f_i w_i(\tau) z_{il}}{\sum_i f_i w_i(\tau)}, \quad (13)$$

with $\bar{x}_{j'}(\tau)$ defined in the same way. The effect at τ is $\hat{\gamma}_j$ plus the heterogeneity coefficients times the composition of the group at τ . Two quantiles differ in the effect only because the groups differ in $\bar{\mathbf{z}}(\tau)$; the profile of $\hat{\theta}_j$ across τ is an exact decomposition into heterogeneity ($\hat{d}_{jl}$, one number per pair) and composition ($\bar{z}_l(\tau)$, one number per quantile). `gepwreg` reports $\bar{z}_l(\tau)$, the population mean $\bar{z}_l$, and the contributions $\hat{d}_{jl}\bar{z}_l(\tau)$ in `e(decomp)`. Evaluated at the population means instead of $\bar{\mathbf{z}}(\tau)$, (13) is the average effect, which is what the interacted regression gives on its own.

The estimator is consistent for $\theta_j(\tau)$ under (11) and $\mathbb{E}[b_{ij} | \mathbf{x}_i, \mathbf{z}_i] = \gamma_j + \sum_l d_{jl} z_{il}$ (plus the own terms): the heterogeneity of the effect runs through $\mathbf{z}$, linearly. That is the price. When the effect varies with something that is ordered with y given $(\mathbf{x}, \mathbf{z})$, the interacted regression averages over it and the profile is flat: that is DGP U1 of Section 8.1, where an irrelevant $\mathbf{z}$ gives the same number at every τ . A flat profile with a weak $\mathbf{z}$ is a symptom, and the second model is the check.

How much of the heterogeneity $\mathbf{z}$ carries. That share is measurable, and measurable before anything is estimated. The law of total variance applied to the coefficient itself splits it into what the model explains and what it misses,

$$\text{Var}(b_{ij}) = \underbrace{\text{Var}(\mathbb{E}[b_{ij} | \mathbf{z}_i])}_{\text{explained}} + \underbrace{\mathbb{E}[\text{Var}(b_{ij} | \mathbf{z}_i)]}_{\text{missed}}, \quad (14)$$

and both terms come from the single fit (11). The first is the dispersion of the fitted $\hat{b}_{ij}$ of (12), a linear combination of the model's own coefficients, so it needs no identification argument and is available for a binary regressor. The second is the coefficient of x_{ij}^2 in the conditional variance of the residual of that fit, since the unexplained part of the effect enters the residual multiplied by x_{ij} . A ratio near zero announces the flat profile: it is 0.002 on U1, against 1.00 on D3 and H1 where $\mathbf{z}$ is the variable that generates the effect.

This is the proportion of random slope variation of the multilevel literature (Rights and Sterba, 2019; Gambino and McCoach, 2025; Rights and Sterba, 2023), read here at the level of the group at τ . Note that no short fit without the interactions is needed, and none would do: (14) is complete as it stands.

4.3 Heterogeneity on the conditional rank: `het(qr)`

The structural model of [Firpo et al. \(2009\)](#) lets the effect vary with the household’s rank in the conditional distribution: $y_i = \mathbf{x}_i' \boldsymbol{\beta}(u_i)$ with $u_i \sim U(0, 1)$ independent of $\mathbf{x}_i$, so that $b_{ij} = \beta_j(u_i)$ and the conditional quantile function is linear in $\mathbf{x}$ at each u . The first step estimates $\boldsymbol{\beta}(u)$ by quantile regression ([Koenker and Bassett, 1978](#)) on a grid $u_1 < \dots < u_G$ ($0.05, 0.10, \dots, 0.95$ by default) and assigns each household its conditional rank: let g_i be the number of grid predictions $\mathbf{x}_i' \hat{\boldsymbol{\beta}}(u_g)$ that lie below y_i , a count that is well defined even where estimated quantile lines cross; then $\hat{u}_i$ is the midpoint of u_{g_i} and u_{g_i+1} , or the end of the grid when $g_i = 0$ or $g_i = G$, and $\hat{b}_{ij}$ is $\hat{\beta}_j$ interpolated linearly at $\hat{u}_i$. The second step is (8). Households with $y_i \approx q_\tau$ have $\hat{u}_i \approx u_\tau(\mathbf{x}_i)$, the conditional rank that puts a household of type $\mathbf{x}_i$ at the population quantile, so the estimator is the UQPE of the structural model, $\mathbb{E}[\omega_\tau(\mathbf{x}) \beta_j(u_\tau(\mathbf{x}))]$, without solving for $u_\tau(\mathbf{x})$.

Relation to [Alejo et al. \(2025\)](#). This estimator is theirs. They show that the unconditional quantile partial effect is the average of the conditional ones taken at a covariate-specific quantile level, $\text{UQPE}_j(\tau) = \mathbb{E}[\beta_j(\xi_\tau(\mathbf{x})) \mid y = q_\tau]$ with $\xi_\tau(\mathbf{x}) = \{\eta : Q_y[\eta \mid \mathbf{x}] = q_\tau\}$ their *matching function*, and estimate it by a Nadaraya–Watson regression of the matched coefficients on the outcome, evaluated at $\hat{q}_\tau$; they establish consistency and asymptotic normality. Three differences remain. Their kernel is in units of y , ours in units of the rank, which removes the scale of the outcome from the bandwidth and is what makes Silverman’s rule and Section 4.10 applicable. Their matching function is evaluated at q_τ for every $\mathbf{x}$, ours at the household’s own y_i ; the two agree to $O(h)$ over the households that carry weight, and ours leaves a household-level $\hat{b}_{ij}$ that is reused for the decomposition and for the initial-quantile object. And the survey design of Section 6.3 is not treated there. The asymptotic theory of this estimator is theirs; `gepwreg` reports the bootstrap for it.

This model captures heterogeneity ordered with y given $\mathbf{x}$ (U1 in the Monte Carlo: 1.18–1.80 recovered) and not heterogeneity that runs through the level of a regressor, because $\boldsymbol{\beta}(u)$ is linear in $\mathbf{x}$ at each u : an effect of household size that depends on household size is a quadratic term that `het(z)` includes and `het(qr)` does not. The two models are two answers to “through what does the effect vary”, and when they disagree the disagreement is informative (Section 9).

4.4 Choosing between the two models

The two heterogeneity models are not interchangeable, and the order in which to consult them is fixed. The share (14) of Section 4.2 is read first, because it comes from the `het(z)` fit alone and needs no comparison. Near one, the observables carry the heterogeneity and `het(z)` is the more precise route by an order of magnitude (Table 10); near zero, they have no purchase and `het(z)` will report a flat profile whatever the data do, which is the U1 failure, visible before any estimate is read. `het(qr)` is then the alternative, and it assumes something different in kind: not that the heterogeneity is explained by observables, but that it is ordered with the household’s rank in the conditional distribution. A necessary condition is that the conditional quantile slopes vary with u ; it is testable, and it is not sufficient, since those slopes move for reasons that need not be a random coefficient at all. When neither holds, neither route identifies the effect, and there are configurations in which $\theta_j(\tau)$ is not defined to begin with, because a random coefficient and a heteroskedastic error are indistinguishable at the second conditional moment and only the first reading gives the

effect a household-level meaning. Those cases, the proofs behind (14), and what a short fit can and cannot see, are in the companion note on the choice of the heterogeneity model.

4.5 Relation to RIF regression

RIF regression (Firpo et al., 2009; Rios-Avila, 2020) regresses $\text{RIF}(y_i; q_\tau) = q_\tau + (\tau - \mathbf{1}[y_i \leq q_\tau])/\hat{f}(q_\tau)$ on $\mathbf{x}_i$. Its population slope is the linear projection of $\mathbb{E}[\text{RIF} | \mathbf{x}] = q_\tau + (\tau - \Pr[y \leq q_\tau | \mathbf{x}])/f(q_\tau)$ on $\mathbf{x}$, and its derivative with respect to x_j is $\omega_\tau(\mathbf{x}) \mathbb{E}[b_{ij} | \mathbf{x}, y = q_\tau]$; averaged, this is (3). RIF regression and the two-step estimator are two representations of the same object. RIF regression models the conditional distribution of y at one point, $\Pr[y \leq q_\tau | \mathbf{x}]$, once per τ , linearly in $\mathbf{x}$ (a projection, exact when $\mathbb{E}[\text{RIF} | \mathbf{x}]$ is linear, otherwise an approximation whose remedies are RIF-logit or a nonparametric first stage), and needs the density $f(q_\tau)$, which is small in the tails. The two-step estimator models the effect once for all τ , uses the ranks for the weights, and needs no density. It pays with the heterogeneity model. Because RIF regression needs no model of the effect, it is the natural check of that model — with the caveat that it is not a neutral arbiter, since its own linear projection fails on exactly the profiles that curve, as Section 8.1 shows: **het**(**z**), **het**(**qr**) and RIF-OLS agree when the heterogeneity is captured, and the direction of a disagreement points to what is missed.

4.6 When RIF regression is exact, and when it is not

The two representations describe the same population quantity, so any disagreement between them comes from what each one has to estimate. That difference can be isolated exactly. Take the simplest model there is,

$$y_i = b x_i + u_i, \quad u_i \perp x_i, \quad b \text{ constant}, \quad (15)$$

with no heterogeneity of the effect, no endogeneity and nothing misspecified. Shifting x by δ shifts every quantile of y by $b\delta$, so the UQPE and $\theta(\tau)$ both equal b at every τ , exactly and with no approximation of any kind. Whatever a RIF regression reports other than b in this design is therefore its own error, with nothing else in the way.

Write $F(q | x) = \Pr[y \leq q | x]$. Since $\mathbb{E}[\mathbf{1}[y \leq q_\tau] | x] = F(q_\tau | x)$, the population slope of a RIF regression is a linear projection of that function,

$$\gamma_\tau^{\text{RIF}} = - \frac{\text{Cov}(F(q_\tau | x), x)}{f(q_\tau) \text{Var}(x)}, \quad (16)$$

while a shift of x by δ moves the conditional distribution by $b\delta$, so $\partial F(q | x)/\partial x = -f(q | x)b$ and the effect is an average derivative of the same function,

$$\theta(\tau) = \frac{\mathbb{E}[-\partial F(q_\tau | x)/\partial x]}{f(q_\tau)} = b. \quad (17)$$

The entire difference between the two estimators, here, is the difference between the linear projection of $F(q_\tau | \cdot)$ and its average derivative.

That difference vanishes in two cases, and only two are easy to name. The first is when $F(q_\tau | \cdot)$ is affine on the support of x , which asks the distribution function of u to be affine over the whole range $q_\tau - b \text{supp}(x)$: u uniform, and no part of the support running off

its edge. It is a knife-edge and of no practical interest. The second is neither. When x is jointly Gaussian, Stein’s lemma gives $\text{Cov}(g(x), x) = \text{Var}(x) \mathbb{E}[g'(x)]$ for any differentiable g , so the projection *is* the average derivative, whatever the law of u . That a least-squares fit on Gaussian regressors recovers an average derivative is classical in the index-model literature (Li and Duan, 1989); what it says here is that a RIF regression is exact when the covariates are Gaussian, and has no reason to be exact otherwise.

Table 4: The departure of each estimator’s population limit from the true effect, in per cent

Law of x	0.05	0.10	0.25	0.50	0.75	0.90	0.95
<i>RIF-OLS, Gaussian error</i>							
normal	0.0	0.0	0.0	0.0	0.0	0.0	0.0
uniform	-8.2	-3.7	2.2	4.8	2.2	-3.7	-8.2
Student t_5	10.6	2.1	-5.5	-8.1	-5.5	2.1	10.6
χ_3^2	-41.7	-36.6	-25.9	-9.5	14.5	46.7	73.2
exponential	-47.2	-42.0	-30.9	-13.1	14.6	54.7	90.2
lognormal	-64.9	-61.1	-52.4	-36.9	-7.8	48.9	119.1
<i>PWR-2, any law</i>	0.0	0.0	0.0	0.0	0.0	0.0	0.0
<i>RIF-OLS, skewed error (lognormal, skewness 4.75)</i>							
normal	0.0	0.0	0.0	0.0	0.0	0.0	0.0
uniform	-39.5	-16.4	20.4	17.5	-11.5	-8.7	-6.4
Student t_5	19.2	-2.1	-16.7	-14.7	-0.5	16.3	23.8
χ_3^2	-71.8	-61.8	-38.8	-3.3	40.0	74.0	81.4
exponential	-78.5	-69.4	-46.7	-8.0	44.0	90.9	106.5
lognormal	-88.0	-83.1	-69.5	-40.6	17.1	117.5	215.9
<i>PWR-2, any law</i>	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Model (15), in which the true effect is b at every τ by construction. Each law is standardised to mean zero and unit variance, and $b \text{sd}(x)/\text{sd}(u) = 0.5$ throughout. Entries are $100(\gamma_\tau^{\text{RIF}}/b - 1)$ for the RIF rows and $100(\theta(\tau)/b - 1)$ for the PWR-2 row, both computed by quadrature from (16) and (17): no sample is drawn, no density is estimated, and the numbers carry no Monte Carlo error. The PWR-2 row is one row rather than six because the six agree to the displayed precision, and PWR-2(z) and PWR-2(qr) coincide here because the effect does not vary; the programme that builds the table prints them separately if they ever do not. File `p10_rif_gap.py`, which also reports the convergence of the Gaussian row.

Table 4 puts numbers on the gap, and the reading is not the one intuition proposes. It is the law of the *regressor*, not the law of the error, that decides. With a Gaussian x the projection is exact to the accuracy of the quadrature even when u has skewness 4.75; with a lognormal x it is out by -65 per cent at $\tau = 0.05$ and $+119$ at $\tau = 0.95$ *when u is Gaussian*, which is the case in which a RIF regression is usually thought safe. A skewed error amplifies what a skewed regressor has already caused — the same row runs from -88 to $+216$ — but it does not cause it alone.

The two-step estimator — PWR-2 of Table 1, not the PWR of Araar (2016, 2023), which is PWR-1 and is the subject of Section 3 — is in the table too, and it is the flat line. In this design the effect does not vary, so the percentile-weighted mean of the unit effects returns b whatever the weights and whatever the two laws are; the estimator has nothing it can get wrong here. That is not a property that survives heterogeneity, and it is not offered as one. The zero is arithmetic, not a claim: a weighted mean of a constant is that constant,

whatever the weights. Once the effect varies, PWR-2 has a smoothing bias of order h^2 , which Section 8.1 measures at 0.006 to 0.011 on a design where its model is right, and a first-order bias when its model is wrong — in the design of that section where the variable carrying the effect is unobserved, PWR-2(z) returns the same number at every quantile for a profile that runs from 1.18 to 1.80, with t up to 60. Section 5 adds a third source, measurement error in the heterogeneity variables, which overstates the effect by sixteen per cent on the design of Section 5.3 before the correction. What the flat line of Table 4 does is not exonerate PWR-2; it removes it from the measurement, so that what the RIF rows show is the projection error and nothing of ours.

Three things follow for practice. The departure changes sign across τ and across laws, so it cannot be signed in advance and no rule of thumb repairs it. A weak model is no protection: the projection and the average derivative do agree in the limit as the signal vanishes, but slowly when x is skewed, and at $\text{bsd}(x)/\text{sd}(u) = 0.02$, where the regressor explains almost nothing, the gap at $\tau = 0.05$ for a lognormal x is still -36 per cent. And what hurts is asymmetry, not thick tails: with a Gaussian error the symmetric laws of the table, uniform and Student t_5 , stay within eleven per cent everywhere, while the three asymmetric ones do not; a skewed error then carries even the symmetric ones beyond that (-40 per cent for the uniform at $\tau = 0.05$). A household survey offers few Gaussian regressors and many asymmetric ones — household size, land, expenditure, age — which is the setting of Section 9.

The variance points the same way, for a separate reason. The recentred influence function has $\text{Var}[\text{RIF}(y; q_\tau)] = \tau(1-\tau)/f(q_\tau)^2$, so the standard error of γ_τ^{RIF} carries a factor $1/f(q_\tau)$ and grows without bound where the density of the outcome is small, and $f(q_\tau)$ is itself estimated at the point where the data are thinnest. The two-step estimator has no such factor. The rank is uniform, so a window of half-width h on the rank holds $2hn$ units at $\tau = 0.05$ exactly as at $\tau = 0.5$: in the coordinate the estimator works in, there is no tail. Which of the two is more precise in a given sample still depends on what the heterogeneity model buys, and Table 13 shows both orderings on the same data; but the factor that can diverge sits on one side only.

None of this says that a RIF regression is wrong. It is the linear projection it advertises, and Firpo et al. (2009) present it as such. What the algebra adds is where that projection is exact — Gaussian covariates — and how far it travels when they are not. It also marks a difference between the two estimands that the equality of Section 4.5 can hide. The UQPE is defined through a marginal location shift, so it needs a covariate that can be shifted; $\theta(\tau) = \mathbb{E}[b \mid y = q_\tau]$ is a descriptive quantity, the mean effect among the units that sit there, and needs no such experiment. It stays defined for the binary regressors — a male head, an urban household — that a survey is largely made of, and for which the derivative in (17) does not exist.

4.7 A stacked-GMM representation

For $\text{het}(\mathbf{z})$ the two-step estimator is an exactly identified GMM estimator. Stack the first-step moment conditions $\mathbb{E}[\mathbf{W}_i e_i] = 0$ (with $\mathbf{W}_i$ the regressors of (11)) and the moment conditions of the group means, $\mathbb{E}[f_i w_i(\tau)(\mathbf{z}_i - \boldsymbol{\mu})] = 0$; the parameter vector is $(\alpha, \boldsymbol{\gamma}, \mathbf{c}, \mathbf{d}, \boldsymbol{\mu})$ and $\theta_j(\tau)$ is the smooth function (13) of it. The GMM point estimate is the two-step estimate, and the GMM sandwich variance of θ_j obtained by the delta method is the influence-function variance of Section 6 when the weights are treated as fixed (Newey

and McFadden, 1994, Section 6); the influence function adds the term that the weights, which depend on the ranks, contribute (Section 6.1). The representation is a check, not a computational route. On a simulated sample of 20,000 households from DGP H1 of Section 8 (file `gepwreg_gmm_check.do`), the GMM point estimates coincide with the closed form to four decimals at $\tau = 0.1, 0.5$ and 0.9 , and the GMM standard errors (0.0086–0.0087) lie between the delta-method standard errors that treat composition and weights as fixed (0.0072–0.0075) and the influence-function standard errors that carry the rank term (0.0088–0.0091), which a pairs bootstrap of the whole procedure at the same bandwidth confirms to within three percent (0.0090–0.0094, on 2,000 replications; the standard error of a bootstrap standard error is about $1/\sqrt{2B}$, so the customary 200 replications cannot resolve a difference of this size).

4.8 The effect at the initial quantile

With household effects in hand, the outcome before the contribution of the regressors in J is $\hat{y}_{0i} = y_i - \sum_{j \in J} \hat{b}_{ij}(x_{ij} - x_j^{\text{ref}})$, and the estimator of Definition 2 is (8) with the weights on the rank of $\hat{y}_0$:

$$\hat{\theta}_{0j}(\tau) = \frac{\sum_i f_i w_{0i}(\tau) \hat{b}_{ij}}{\sum_i f_i w_{0i}(\tau)}, \quad w_{0i}(\tau) = K\left(\frac{\hat{F}_{0n}(\hat{y}_{0i}) - \tau}{h_0}\right). \quad (18)$$

The households are re-ranked, so the composition of the group changes and with it the effect. The reference value is the level of x_j from which the contribution is counted, zero by default (`xref()` sets another); for a programme indicator y_0 is then the outcome without the programme. Standard errors are by the bootstrap of both steps, because the re-ranking is a discontinuous function of the first-step estimates. `gepwreg` reports $\hat{\theta}_{0j}$ as the coefficient and keeps $\hat{\theta}_j(\tau)$, the effect at the quantile of the observed outcome, in `e(b_atq)`.

4.9 The profile along a covariate

The estimator of Definition 3 is the one-step regression (5) with the weights on the rank of z instead of y :

$$\hat{\mathbf{b}}_z(\tau) = (\mathbf{X}'\mathbf{W}_z\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}_z\mathbf{y}, \quad \mathbf{W}_z = \text{diag}\{f_i K((\hat{F}_n^z(z_i) - \tau)/h)\}. \quad (19)$$

It is a locally constant varying-coefficient regression in z : among households with $z_i \approx q_\tau^z$, y varies with $\mathbf{x}$, and the weighted slope converges to $\mathbb{E}[b_{ij} | z_i = q_\tau^z]$ when the household effects are mean independent of $\mathbf{x}$ given z , with the usual $O(h^2)$ smoothing bias in z , for which the MSE-optimal rule of the earlier paper, the default, applies. Everything derived for this estimator before — its influence function, its survey-design variance — is unchanged; Section 8 adds its coverage.

4.10 The bandwidth of the second step

The second step (8) is a Nadaraya–Watson regression of a household-level quantity on the rank of the outcome, evaluated at τ . The rank is uniform on $[0, 1]$ by construction, so the design density is flat and the usual $g'f'/f$ term of the local constant bias vanishes: only the curvature of the profile survives. With $\sigma_K^2 = 2$ and $R(K) = \int K^2 / (\int K)^2 = 1/(2\sqrt{2\pi})$ for the kernel (4), the asymptotic mean squared error of $\hat{\theta}_j(\tau)$ is

$$\text{MSE}(h) = h^4 \theta_j''(\tau)^2 + \frac{s_j^2(\tau)}{2\sqrt{2\pi} n h}, \quad s_j^2(\tau) = \text{Var}(b_{ij} | p_i = \tau), \quad (20)$$

minimised at

$$h_j^*(\tau) = \left[\frac{s_j^2(\tau)}{8\sqrt{2\pi} n \theta_j''(\tau)^2} \right]^{1/5}. \quad (21)$$

This is the MSE-optimal bandwidth of the earlier paper, with the curvature of the profile in place of β'' and the dispersion of the household effects within the group in place of the local residual variance.

One call reports k coefficients and must use one bandwidth, so (21) has to be aggregated. The coefficients do not share units, and a rule that minimised $\sum_j \text{MSE}_j$ would be decided by whichever regressor happens to be measured in the largest units. Minimising the average *relative* mean squared error, $k^{-1} \sum_j \text{MSE}_j(h)/s_j^2$, removes the scale and gives

$$h^*(\tau) = \left[\frac{k}{8\sqrt{2\pi} n \sum_{j=1}^k (\theta_j''(\tau)/s_j(\tau))^2} \right]^{1/5}, \quad (22)$$

which reduces to (21) when $k = 1$ and is invariant to rescaling any x_j . The plug-in needs a pilot, taken at $2h_{\text{Sil}}$: θ_j'' by second differences on a grid of step $d = \min(0.05, \tau/2, (1-\tau)/2)$, s_j^2 as the weighted variance of $\hat{b}_{ij}$ around $\hat{\theta}_j$, then (22) clipped to $[0.5, 3] h_{\text{Sil}}$. It is cheap: the household effects $\hat{b}_{ij}$ of (13) are produced by the first step and do not depend on h at all, so the pilot is three weighted means over quantities already in memory — no refitting, and no separate pilot regression.

`gepwreg` 1.4 implements (22) and makes it the default for every estimator, one-step and two-step alike. The option `silverman` restores the rank-scale Silverman rule and `band(#)` fixes h by hand.

Three qualifications. The estimation error of the first step contributes a variance that does not depend on h — for `het(z)`, $\hat{\gamma}_j$ and $\hat{d}_{jl}$ in (13) — so the trade-off (20) governs only the composition part, the $\bar{z}_l(\tau)$. The plug-in inherits the instability of the earlier rule when the profile is locally straight: $\mathbb{E}[\hat{\theta}_j''] = \theta_j''^2 + \text{Var}(\hat{\theta}_j'') > 0$, so the denominator of (21) is inflated and h^* is pulled *down*, not up, which is what the clipping is for. And $[0, 1]$ has boundaries: at τ near 0 or 1 the kernel is truncated and the bias is $O(h)$ rather than $O(h^2)$, so (21) no longer describes it.

What the rule is worth. Table 5 measures how much of the optimum the plug-in recovers from one sample, on the DGPs of Section 8.1 at three sample sizes, against the exact $\theta_j(\tau)$ and against the *oracle* bandwidth — the one that minimises the realised root mean squared error in each cell, which no rule can attain. The summary is each cell’s root mean squared error divided by the oracle’s, not the average of the errors themselves: the cells differ in scale by a factor of fifty, so averaging levels would let one DGP decide the comparison.

The plug-in cuts the average distance to the oracle from 33 to 13 percent and the worst cell from 216 to 52 percent. Where Silverman’s rule fails it fails badly, and in a case that is neither exotic nor rare: D1 at $\tau = 0.1$ and 0.9 is a steep, nearly straight profile at the boundary of the rank, where the local constant average is almost entirely bias — 0.0087 of a root mean squared error of 0.0092 at $n = 6,000$ — and the plug-in halves the bandwidth and removes it. Silverman’s rule is a density rule; it knows nothing about the curvature of a profile, and at a boundary that ignorance is expensive.

The rule has one configuration it does not handle, and the user should know which: τ close to a boundary of the rank with a profile that bends there. On H1 at $\tau = 0.9$ the plug-in sits at 1.4 to 1.5 times the oracle against 1.0 to 1.4 for Silverman’s rule, at all three sample sizes. The cause is the boundary acting on the pilot rather than on the estimate: the second difference reads $\hat{\theta}$ at $\tau + d$, where the pilot kernel is cut by the support, so the curvature is underestimated and h^* comes out too wide. `silverman` is the fallback there, and Section 6.2 gives a second reason to take it.

Table 5: The plug-in bandwidth against the Silverman rule and against the oracle

	Mean	Median	Worst	Cells more than 10% off
Silverman’s rule	1.326	1.096	3.161	22/45
Plug-in (22)	1.131	1.063	1.524	17/45

Root mean squared error relative to the oracle bandwidth, over 45 cells: DGPs D3, H1 and D1 of Table 10, $\tau \in \{0.1, 0.25, 0.5, 0.75, 0.9\}$, $n \in \{2,000, 6,000, 20,000\}$, 300 replications, against the exact $\theta(\tau)$. The oracle minimises the realised root mean squared error in each cell over a grid of h from 0.4 to $3h_{\text{Sil}}$, so 1.000 is unattainable by any rule that must choose from one sample. U1 is excluded: its heterogeneity variable is irrelevant by construction, the bias is 0.2–0.3 at every bandwidth, and no bandwidth rule can be judged on it. File `pwr_bandwidth_check.py`.

5 Measurement error in the heterogeneity variables

The heterogeneity of an effect, in survey data, is carried by variables that are rarely measured well: household size, years of schooling, a land area, an expenditure aggregate standing in for living standards. This section asks what error in $\mathbf{z}$ does to the profile, gives a correction that the sample itself identifies, and states the two things that correction cannot do.

Write $\tilde{z}_{il} = z_{il} + u_{il}$, with u_{il} classical: mean zero, symmetric, and independent of the true value, of the regressors and of the error of (11). Only `het(varlist)` is at stake. The correction is a subtraction on a matrix of second moments, which least squares provides and the check function of `het(qr)` does not.

5.1 Where the damage falls

Not where the textbook attenuation would put it.

The profile (13) has two ingredients: the first-step coefficients, and the composition $\bar{z}_l(\tau)$ of the group at τ . The composition survives. The weights are a function of the rank of y , and y is generated by the true $\mathbf{z}$, so u is independent of that rank; the weighted mean of $\tilde{z}_l$ over the group at τ therefore estimates $\bar{z}_l(\tau)$ without bias, at every quantile. The error does not move the group, and the decomposition of the profile into composition and heterogeneity is untouched.

The whole of it falls on the first step. Let $\mathbf{W}_i$ be the regressor vector of (11) and $\tilde{\mathbf{W}}_i$ the same with $\tilde{\mathbf{z}}_i$ in place of $\mathbf{z}_i$. Because u is independent of y , the cross moments with the outcome are unchanged, $\mathbb{E}[\tilde{\mathbf{W}}_i y_i] = \mathbb{E}[\mathbf{W}_i y_i] \equiv \mathbf{m}_y$, while the second moments are inflated in every block that involves $\mathbf{z}$,

$$\mathbb{E}[\tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i'] = \mathbb{E}[\mathbf{W}_i \mathbf{W}_i'] + \mathbf{\Omega}(\sigma^2) \equiv \mathbf{M} + \mathbf{\Omega}(\sigma^2), \quad (23)$$

with $\mathbf{\Omega}$ linear in the error variances $\sigma_l^2 = \text{Var}(u_{il})$ and zero elsewhere. The naive fit returns $\tilde{\mathbf{b}} = (\mathbf{M} + \mathbf{\Omega})^{-1} \mathbf{m}_y$ in place of $\mathbf{b} = \mathbf{M}^{-1} \mathbf{m}_y$, so that

$$\tilde{\mathbf{b}} - \mathbf{b} = -(\mathbf{M} + \mathbf{\Omega})^{-1} \mathbf{\Omega} \mathbf{b}. \quad (24)$$

Equation (24) says two things, and the second is the one that matters here. Since $\mathbf{\Omega}$ is not diagonal and $\mathbf{M}$ is not block diagonal, the bias does not stay with the coefficients of the mismeasured variables: it reaches $\hat{\gamma}_j$, the coefficient of a regressor that is measured exactly. In (13) $\hat{\gamma}_j$ is the one term that does not depend on τ . Error in $\mathbf{z}$ therefore shifts the *level* of the profile as well as shrinking its slope in $\bar{\mathbf{z}}(\tau)$, and the level is where most of the reported effect sits. A reader who inspects the interaction coefficients $\hat{d}_{jl}$, finds them plausible, and reads the profile as though only they mattered will be reading a level that is wrong. The estimator of Section 4.2 is thus more exposed than the interacted regression it is built on, whose interaction coefficients are the usual object of scrutiny.

5.2 Identification from third moments

The correction is (23) read backwards: recover σ^2 and solve $(\mathbf{M} + \mathbf{\Omega} - \mathbf{\Omega})\mathbf{b} = \mathbf{m}_y$, that is, subtract $\mathbf{\Omega}(\hat{\sigma}^2)$ from the observed moment matrix before inverting it. Everything turns on estimating σ^2 without an instrument and without a validation sample.

The first-step residual carries the information. With $\varepsilon_i = y_i - \tilde{\mathbf{W}}_i' \mathbf{b}$, the residual at the coefficients of the correctly measured model, the error enters ε multiplied by the coefficients it contaminates, and ε is correlated with the *square* of the mismeasured variable in a way that depends on σ_l^2 alone. For a symmetric error,

$$\mathbb{E}[\varepsilon_i \tilde{z}_{il}^2] = -2\sigma_l^2 \left(c_l \mathbb{E}[z_{il}] + \sum_j d_{jl} \mathbb{E}[x_{ij} z_{il}] \right), \quad (25)$$

$$\mathbb{E}[\varepsilon_i x_{ij} \tilde{z}_{il}^2] = -2\sigma_l^2 \left(c_l \mathbb{E}[x_{ij} z_{il}] + \sum_{j'} d_{j'l} \mathbb{E}[x_{ij} x_{ij'} z_{il}] \right), \quad j = 1, \dots, k, \quad (26)$$

and every moment on the right is estimable, because u drops out of each: $\mathbb{E}[\tilde{z}_l] = \mathbb{E}[z_l]$ and $\mathbb{E}[x_j \tilde{z}_l] = \mathbb{E}[x_j z_l]$. One error variance is therefore over-identified by $k + 1$ conditions. The cross terms between two heterogeneity variables vanish, so the system separates: one σ_l^2 per variable, each with its own $k + 1$ moments. `gepwwreg` solves it by GMM with the efficient weight matrix, in two passes, and the standard error of $\hat{\sigma}_l^2$ comes from the same criterion. The residual is evaluated at the corrected coefficients $\mathbf{b}(\sigma^2) = (\mathbb{E}[\tilde{\mathbf{W}}_i \tilde{\mathbf{W}}_i'] - \mathbf{\Omega}(\sigma^2))^{-1} \mathbf{m}_y$, so that σ^2 and $\mathbf{b}$ are estimated jointly; at the naive coefficients $\tilde{\mathbf{b}}$, (25)–(26) would carry the extra term $\mathbb{E}[\tilde{z}_{il}^2 \tilde{\mathbf{W}}_i']'(\mathbf{b} - \tilde{\mathbf{b}})$.

Two conditions are necessary, and neither is sufficient. The bracket on the right of (25)–(26) must not vanish: the heterogeneity variable must have a non-zero mean or be correlated with the regressors, and its coefficients in the first step must not be zero. And the latent z must be non-Gaussian, since with everything Gaussian the third-moment structure that carries σ_l^2 collapses. A skewed variable correlated with the regressors is the favourable case; a symmetric variable independent of them carries almost no information about its own error, however large that error is.

Skewness is a usable signal because measurement error dilutes it and never creates it: with symmetric error, $\text{skew}(\tilde{z}) = \text{skew}(z) \lambda^{3/2}$, where $\lambda = \text{Var}(z)/\text{Var}(\tilde{z})$ is the reliability. The skewness one can see is a lower bound on the skewness one cannot, so a proxy that still looks skewed is evidence that the latent variable is more so. No distributional assumption is needed for the correction to be valid; the shape governs only its precision.

5.3 When it applies, and when it abstains

Necessary conditions are not enough to license a correction on a given sample, so the decision is made from the data. For each heterogeneity variable `gepwreg` forms $t = \hat{\sigma}_l^2 / \text{se}(\hat{\sigma}_l^2)$ and corrects only where t exceeds a threshold, two by default (`tcrit`). Where the variance is not identified, where the maintained assumptions fail, or where there is no error at all, $\hat{\sigma}_l^2$ goes to zero and the estimator falls back on the uncorrected fit. The decision is made once per variable, not once per quantile, since the first step does not depend on τ .

A simulation makes the size of the stakes concrete; it verifies (24) rather than establishing it. Two heterogeneity variables, z_1 skewed and correlated with the regressor, z_2 symmetric and independent of it, both observed with error, $n = 8,000$. At $\tau = 0.75$ the true effect is 2.976. Run on the true variables the estimator returns 2.963; run on the proxies it returns 3.453, over-stated by 16 per cent, while the interaction coefficients remain unremarkable. The correction estimates 0.249 for the first error variance, whose true value is 0.25, with $t = 19$, and abstains on the second, whose t is 1.3: z_2 is symmetric and independent of the regressor, so its error is not identified however large it is — and here its reliability is 0.57, which is to say the error is not small. The corrected effect is 2.956, within 0.7 per cent of the truth.

The correction removes a bias and pays for it in variance, and the price is not negligible: as the reliability falls, the corrected profile stays centred on the truth while its dispersion grows. At the threshold itself the estimator is a pre-test estimator, and no standard error is uniformly valid there. Where t is close to `tcrit`, the corrected and the uncorrected profile should be reported side by side rather than the corrected one alone.

5.4 What is not identified

Three cases, and the first is the one that looks most innocent.

A heterogeneity variable taking two values — an indicator of a factor variable included — carries no information about its own error. If z takes only a and b then $z^2 = (a+b)z - ab$ is affine in z , so (25)–(26) are linear combinations of moments that the first step has already set to zero. Centring does not help, $(z-c)^2$ being affine as well. The variance is not weakly identified, it is not identified at all, and a t computed from it is the ratio of two numerical residuals. Error in a categorical variable is misclassification in any case, mechanically correlated with the true value, which is not the object corrected here.

The square of a mismeasured variable does not carry classical error either: $\tilde{z}^2 = z^2 + 2zu + u^2$, whose error is correlated with the true value. Entering a categorical code as though it were continuous is the usual way into that trap.

Finally, error in the regressors themselves is outside the scope. If a heterogeneity variable is also one of the $\mathbf{x}$, the error reaches the regressor block, and that is the case where the effect of interest is not defined.

5.5 The limit no test removes

The identifying moments cannot tell a missing functional form from measurement error. The claim is algebraic: $\mathbb{E}[\varepsilon \tilde{z}^2]$ can be non-zero because b_{ij} is nonlinear in z exactly as much as because z carries noise, and (25)–(26) do not separate the two causes. If the effect is

not linear in the heterogeneity variable, the curvature lands in the first-step residual and the third moments read it as an error variance.

On a variable measured exactly whose effect is quadratic — so that the true error variance is zero — the correction fires with $t = 11.4$, and moves the profile part of the way towards what the correctly specified model gives with no correction at all: at $\tau = 0.75$ the truth is 2.714, the linear uncorrected fit gives 3.027, the linear fit with the correction 2.832, and the quadratic fit without any correction 2.716.

The limit is real and no diagnostic internal to the correction removes it. One comparison reads it, and it costs a single command. Fit the enriched model — a square, or `i.` for a categorical code — *without* the correction, and see where it puts the profile. If it lands where the correction put it, the correction was doing the work of the missing term. If it stays where the uncorrected linear model was, the correction is answering something no functional form will answer. The two cases are mirror images, and the simulation of Section 5.3 is the second one: the enriched model without the correction returns 3.345, close to the uncorrected 3.453 and far from the corrected 2.956.

5.6 Inference

The influence function of Section 6.1 accounts for the estimation of σ^2 : the score of the moment conditions (25)–(26) enters the influence function of $\hat{\theta}_j(\tau)$ alongside the score of the first step, so (29) and its survey-design counterpart remain valid in the regime where the correction is active. The bootstrap re-runs the whole procedure, the estimation of σ^2 and the threshold included, at every replication. The pre-test caveat of Section 5.3 is the exception, and it is a statement about the neighbourhood of the threshold, not about the corrected estimator away from it.

Relation to the literature. Identifying the variance of a classical measurement error from higher moments of the data, without instruments and without a validation sample, is an old idea, and it has been developed for the errors-in-variables model by Erickson and Whited (2002) in a GMM framework and, for additive and polynomial models of a mismeasured regressor, by Ben-Moshe et al. (2017). The structural form of (11) — linear in the regressors, linear in the heterogeneity variables, with their products — is of the polynomial class those authors identify. What is specific here is neither the idea nor the class. It is the place the conditions occupy: (25)–(26) are unconditional moments of a least-squares residual, one for the constant and one for each regressor, separable across heterogeneity variables, where Ben-Moshe et al. (2017) identify their functions from conditional covariances given the correctly measured regressor and require the latent mismeasured regressor to be independent of it once a conditional mean is removed — a restriction (11) does not impose. And it is the object: the target here is not the structural function but $\theta_j(\tau)$, a weighted average of derived household effects, whose bias under measurement error is the level shift of Section 5.1 rather than the attenuation of a single coefficient, and whose influence function must carry $\hat{\sigma}^2$ through the averaging and through the estimated ranks.

6 Inference

6.1 Influence function of the two-step estimator on observables

Write the closed form (13) as $\theta_j = \gamma_j + \sum_l d_{jl}\mu_l$ (the own terms add the same structure with $\bar{x}_{j'}(\tau)$ and are omitted from the notation), with $\boldsymbol{\mu}(\tau) = \bar{\mathbf{z}}(\tau)$. Its influence function is

$$\psi_i(\theta_j) = \psi_i(\gamma_j) + \sum_{l=1}^m [\psi_i(d_{jl})\mu_l + d_{jl}\psi_i(\mu_l)], \quad (27)$$

where $\psi_i(\gamma_j)$ and $\psi_i(d_{jl})$ are the OLS influence functions of the first step, the rows of $(\mathbf{W}_i f_i e_i)' \mathbf{A}^{-1}$ with $\mathbf{A} = n^{-1} \sum_i f_i \mathbf{W}_i \mathbf{W}_i'$, and $\psi_i(\mu_l)$ is the influence function of a percentile-weighted mean. The weights depend on the ranks, and a household i enters μ_l twice: directly through its own weight, and indirectly because its outcome shifts the rank of every household with $y_{i'} \geq y_i$ by $1/n$, and with it their weights. With $\dot{w}_{i'} = \partial w_{i'}/\partial p_{i'} = -(p_{i'} - \tau)w_{i'}/(2h^2)$ and $a_{i'l} = \dot{w}_{i'} f_{i'}(z_{i'l} - \mu_l)$,

$$\psi_i(\mu_l) = \frac{1}{S} \left[n f_i w_i (z_{il} - \mu_l) + f_i \sum_{i'} a_{i'l} (\mathbf{1}[y_{i'} \geq y_i] - p_{i'}) \right], \quad S = \sum_i f_i w_i, \quad (28)$$

which is the score of the earlier paper with $\mathbf{x}e$ replaced by $(\mathbf{z} - \boldsymbol{\mu})$ and is computed in $O(n \log n)$ by a reverse cumulative sum after sorting on y (Appendix A). The variance estimator is

$$\hat{V}_{\text{IF}}(\theta_j) = \frac{1}{n^2} \sum_i \tilde{\psi}_i(\theta_j)^2, \quad \tilde{\psi}_i = \psi_i - \bar{\psi}, \quad (29)$$

with the scores centred. Under the conditions of Proposition 3 and standard regularity for the first step, $(\hat{\theta}_j - \theta_j)/\hat{V}_{\text{IF}}(\theta_j)^{1/2}$ is asymptotically standard normal. The rate is that of a kernel average, not $\sqrt{n}$: the composition term $\hat{\mu}_l(\tau)$ averages over a window of width h , so (29) is of order $1/(nh)$, while the first-step terms are of order $1/n$. The smoothing term of (9) is negligible against the standard error when $nh^5 \rightarrow 0$; at the MSE-optimal bandwidth of Section 4.10 the two are of the same order, which is what an MSE-optimal choice means. In the interior of the rank the smoothing term is small at the sample sizes of household surveys; near a boundary it is not, and Section 8.2 measures it (Table 11, the coverage against the Monte Carlo mean and against the object at $\tau = 0.9$). The influence function of the effect at the initial quantile is not derived, because the re-ranking on $\hat{y}_0$ is not a smooth function of the first-step estimates; the bootstrap is used.

6.2 The bandwidth is chosen, and (29) conditions on it

Everything above holds with h fixed. Since (22) is the default, h is not fixed: it is $\hat{h}^*$, a function of the same sample. To first order,

$$\hat{\theta}_j(\hat{h}^*) - \hat{\theta}_j(\bar{h}) \approx \frac{\partial \hat{\theta}_j}{\partial h}(\bar{h}) (\hat{h}^* - \bar{h}), \quad (30)$$

so the variance of the reported estimate exceeds (29) by $(\partial \theta_j / \partial h)^2 \text{Var}(\hat{h}^*)$, a term (29) cannot contain because it conditions on h .

It is negligible wherever $\partial \theta_j / \partial h$ is small, which is wherever the bandwidth does not matter much — and that is most of the range, because $\partial \theta_j / \partial h$ is $O(h)$ in the interior and the root

mean squared error curve is flat near its minimum. It is not negligible where that curve is steep, which is the boundary of the rank with a curved profile: exactly the configuration in which the plug-in is least reliable to begin with (Section 4.10). Section 8.2 measures the term and its consequence for coverage.

Three responses are available, and `gepwreg` supports all three. The bootstrap re-chooses the bandwidth inside every resample, so it prices (30) automatically, and Section 8.2 verifies that it does: it is the recommended standard error at τ near 0 or 1, and whenever `e(h)` comes back near the top of its clipping range. `silverman` makes the bandwidth a function of the ranks alone, which are uniform by construction, hence effectively deterministic, and restores (29) exactly — at the cost of the point estimate (Table 5). `band(#)` does the same by fiat when the bandwidth is set from outside the data. What is not available is a free lunch: a data-driven bandwidth and a fixed- h variance cannot both be correct, and the combination is the one configuration to avoid reporting without a check.

6.3 Complex survey design

Under stratified cluster sampling with strata $s = 1, \dots, H$ and n_s primary sampling units in stratum s , the Taylor linearisation variance sums the scores within PSUs and takes their between-PSU variance within strata (Rao, 1979; Deville, 1999):

$$\hat{V}_{\text{svy}}(\theta_j) = \frac{1}{n^2} \sum_{s=1}^H \frac{n_s}{n_s - 1} \sum_{c=1}^{n_s} (\Psi_{sc} - \bar{\Psi}_s)^2, \quad \Psi_{sc} = \sum_{i \in (s,c)} \tilde{\psi}_i(\theta_j). \quad (31)$$

This is what `svy:` would do to a fixed score; the score here is (27), which carries the first step and the ranks. `gepwreg` reads the design from `svyset` and reports (31) by default when a design is declared (`vce(if)` requests (29) instead). The stratified cluster bootstrap resamples PSUs within strata and re-runs both steps, including the ranks; it is the reference for the Taylor variance in the Monte Carlo.

6.4 Standard errors side by side

There is one correct variance for each estimator, and practitioners are shown one: the influence-function variance, or its Taylor linearisation when a design is declared, for `het(z)` and `rankvar`; the bootstrap for `het(qr)` and `initial`. The post-estimation command `gepwreg_setable` (also the option `setable`) lists the influence-function, Taylor and bootstrap standard errors of the same estimates with their ratios, as a check of the asymptotic approximation on a given sample, not as a menu. The naive weighted-least-squares variance of the one-step estimator, which treats the weights as fixed and which version 2 showed to be inconsistent, is kept for the replication of that result (`gepwreg_setable, naive`) and is otherwise not displayed.

7 The profile across quantiles: the group and a given household

The profile $\tau \mapsto \theta_j(\tau)$ is the usual way to read heterogeneity along the distribution: an effect that is larger at the top than at the bottom. The group at τ is defined by the outcome, however, and it changes with τ ; a household is at the bottom in part because of

its covariates. Conditioning (3) on the observables of the heterogeneity model as well as on $\mathbf{x}$,

$$\theta_j(\tau) = \mathbb{E}[\omega_\tau(\mathbf{x}_i, \mathbf{z}_i) \theta_j^{\mathbf{x}_i, \mathbf{z}_i}(\tau)], \quad \theta_j^{\mathbf{x}, \mathbf{z}}(\tau) = \mathbb{E}[b_{ij} \mid \mathbf{x}_i = \mathbf{x}, \mathbf{z}_i = \mathbf{z}, y_i = q_\tau], \quad (32)$$

with $\omega_\tau(\mathbf{x}, \mathbf{z}) = f(q_\tau \mid \mathbf{x}, \mathbf{z})/f(q_\tau)$. The profile moves for two reasons. The weights move, because the households found at q_τ are not the same at every τ : that is the composition of the group. And the type-specific effect $\theta_j^{\mathbf{x}, \mathbf{z}}(\tau)$ moves, because a household of a given type need not have the same effect at every quantile: that is the heterogeneity of the effect along the distribution, the only one of the two that says how the effect of x_j varies with the position of a household. This section separates them, says what each heterogeneity model can say about each, and gives the inference for the whole profile.

7.1 The group, a given household, and the average

Definition 4 (Effect for a given household). For a type $(\mathbf{x}^\circ, \mathbf{z}^\circ)$, the reference household, $\theta_j^\circ(\tau) = \mathbb{E}[b_{ij} \mid \mathbf{x}_i = \mathbf{x}^\circ, \mathbf{z}_i = \mathbf{z}^\circ, y_i = q_\tau]$, the expected effect of x_j for the households of that type found at the τ -quantile of the outcome.

Three numbers can be reported at a quantile: the effect for the group, $\theta_j(\tau)$; the effect for a given household, $\theta_j^\circ(\tau)$; and, with no reference to τ , the average effect $\mathbb{E}[b_{ij}]$. The profile of the second holds the type fixed, so its variation across τ is the heterogeneity along the distribution and nothing else, and the gap between the first two at a given τ is the composition of the group. What can be said about each depends on what the heterogeneity model lets the effect depend on (Table 6).

Observables. Under $\text{het}(\mathbf{z})$ the household effect (12) is a function of the observables only, $b_{ij} = b_j(\mathbf{x}_i, \mathbf{z}_i)$, affine in them. A household of a given type has the same effect wherever it is found: $\theta_j^\circ(\tau) = b_j(\mathbf{x}^\circ, \mathbf{z}^\circ)$ at every τ , and the curve of the reference household is flat by construction. The effect for the group at τ is the effect of the mean household of that group, $b_j(\bar{\mathbf{x}}(\tau), \bar{\mathbf{z}}(\tau))$, which is the closed form (13): the profile follows the composition of the group in the heterogeneity variables, and nothing else. Because b_j is affine, the mean household of the population is also the average effect, $b_j(\mathbb{E}\mathbf{x}, \mathbb{E}\mathbf{z}) = \mathbb{E}[b_{ij}]$. Under this model a profile that moves says that the composition moves; whether the effect of a given household changes with its position is a question the model does not ask.

Conditional rank. Under $\text{het}(\mathbf{qr})$, $b_{ij} = \beta_j(u_i)$, and a household of type $\mathbf{x}$ found at q_τ has the conditional rank $u_\tau(\mathbf{x})$ that solves $\tilde{\mathbf{x}}'\boldsymbol{\beta}(u) = q_\tau$, with $\tilde{\mathbf{x}} = (\mathbf{x}', 1)'$: the matching function of Alejo et al. (2025). The effect for a given household is therefore

$$\theta_j^\circ(\tau) = \beta_j(u_\tau(\mathbf{x}^\circ)), \quad (33)$$

which moves with τ through the conditional rank only: for a given type, a higher quantile of the outcome is a higher conditional rank. The profile of the group, $\mathbb{E}[\omega_\tau(\mathbf{x}) \beta_j(u_\tau(\mathbf{x}))]$, mixes the two channels; the curve of the reference household isolates the second. The model lets the effect vary with the rank and not with the level of a regressor (Section 4.3), so the two models answer different questions about the same profile, and a disagreement between them is read through what each lets the effect depend on.

RIF regression and conditional quantile regression. RIF regression estimates $\theta_j(\tau)$ without a model of the household effect, so it has no curve for a given household: its profile is the profile of the group, composition included, and nothing in it separates the two channels. Conditional quantile regression estimates $\beta(u)$, which is a statement about a given household, but indexed by its rank in its own conditional distribution rather than by the quantile of the outcome at which it is found. The profile of $\beta_j(u)$ across u and the curve $\theta_j^\circ(\tau)$ across τ are the same function read on two axes, and the change of axis, $u_\tau(\mathbf{x}^\circ)$, depends on the type: (33) is that change of axis.

Table 6: What each approach lets the effect depend on, and what its profile across quantiles shows

	The effect may depend on	The profile of the group, $\theta_j(\tau)$	The curve of a given household, $\theta_j^\circ(\tau)$
RIF-OLS	— (no model of the household effect)	composition and heterogeneity, not separated	not available
Conditional QR	the conditional rank u	not estimated directly	$\beta_j(u)$, on the axis of u , not of τ
het (z)	the observables z , linearly	composition only	flat: $b_j(\mathbf{x}^\circ, \mathbf{z}^\circ)$
het (qr)	the conditional rank u	composition and the rank	$\beta_j(u_\tau(\mathbf{x}^\circ))$, (33)

Separating composition from structure is the subject of the decomposition literature, between two distributions: the reweighting of DiNardo et al. (1996), the counterfactual distributions of Chernozhukov et al. (2013), the RIF decompositions of Firpo et al. (2018) and the decomposition of the composition effect itself by Rothe (2015); Rothe (2012) changes the marginal distribution of one covariate and holds the dependence between covariates fixed. Here the two groups are the households at two quantiles of the same outcome, and the counterfactual is not a reweighted distribution but a type held fixed along τ .

7.2 The effect for a given household

The reference type is chosen by the user, and `gepwreg` offers three choices (option `ref()`). The default is a type that exists in the data: each continuous variable at its weighted mean, each binary variable at its most frequent value and each factor at its most frequent level. The mean household in the strict sense — every variable at its mean, a binary one at its share and each indicator of a factor at the share of its level, `ref(mean)` — is the average effect under `het(z)`; it is not a type that exists when a regressor is binary or categorical, and under `het(qr)` it evaluates the model at a household that cannot be observed. Values can also be given.

Under `het(z)` the estimator is (12) at the reference, $\hat{\theta}_j^\circ = b_j(\mathbf{x}^\circ, \mathbf{z}^\circ)$, a linear combination of the first-step coefficients that depends neither on τ nor on a bandwidth. Its influence function is the same combination of the first-step influence functions of Section 6.1, plus, when a reference value is estimated (a mean or a share), the influence function of that mean times the derivative of b_j with respect to it. It is exact, and its Taylor linearisation under a design is (31).

Under **het(qr)** the estimator follows (33) on the grid of quantile regressions. Let $\hat{Q}_g^\circ = \tilde{\mathbf{x}}' \hat{\beta}(u_g)$ be the conditional quantiles of the reference household at the grid points, $\hat{q}_\tau$ the weighted empirical τ -quantile of y , and g the segment with $\hat{Q}_g^\circ \leq \hat{q}_\tau < \hat{Q}_{g+1}^\circ$. Then

$$t = \frac{\hat{q}_\tau - \hat{Q}_g^\circ}{\hat{Q}_{g+1}^\circ - \hat{Q}_g^\circ}, \quad \hat{u}_\tau^\circ = u_g + t(u_{g+1} - u_g), \quad \hat{\theta}_j^\circ(\tau) = (1 - t)\hat{\beta}_j(u_g) + t\hat{\beta}_j(u_{g+1}) : \quad (34)$$

the root of $\tilde{\mathbf{x}}' \hat{\beta}(u) = \hat{q}_\tau$ on the piecewise-linear path of the grid, and the coefficients there. It is consistent for (33) under the model of Section 4.3, with a grid fine enough for the interpolation to be negligible, and **gepwreg** reports $\hat{u}_\tau^\circ$. Where $\hat{q}_\tau$ lies outside $[\hat{Q}_1^\circ, \hat{Q}_G^\circ]$, the reference household would need a conditional rank beyond the grid to be found at q_τ , and the estimate is held at the end of the grid; the command says at which quantiles. The curve is not smooth: it is one point of the piecewise-linear $\hat{\beta}(u)$, which carries the sampling noise of each quantile regression and changes slope at every grid point, where the profile of the group averages over all the households near τ .

7.3 Which object for which question

The three objects answer different questions, and the reference type is the analyst's choice (Table 7). The effect for the group, $\theta_j(\tau)$, is the answer for the households that are at τ as they are, their characteristics included: by (2) it is what a change of x_j for every household does to the τ -quantile, the object of a universal policy and of any indicator defined at a quantile, such as a poverty line at the first quartile. That it carries the composition of the group is intended there: the policy reaches the households that are at τ . The mean household, under **het(z)**, is the average effect in the population, an aggregate benchmark; its gap to $\theta_j(\tau)$ is the composition term of (13), the part of the effect at τ that is due to who is there. A type that exists states the result for a household the reader can picture — in the application, a rural household of average size headed by a man who farms for subsistence — and under **het(qr)** its curve answers whether the effect of a given household changes with its position in the distribution. Values given target a group: the type a programme is aimed at; a region, when the region enters the model as a factor, with one curve per region and the rest of the type held fixed, which separates what is due to the region from what is due to who lives there; or a typology of households, whose effects at their positions feed a simulation of a marginal change, the effect at the initial quantile (Definition 2) taking over for a change large enough to re-rank the households.

All of them are partial effects of a marginal change, with no general-equilibrium response, and they hold under the heterogeneity model chosen. Under **het(z)** the effect varies only with the heterogeneity variables, so a curve that is flat for every type is the model's statement and not a finding. Under **het(qr)** it varies only with the conditional rank, and a type that would need a rank beyond the grid has its curve held at the end of it. A reference outside the range of the data extrapolates the model, which is one more reason to prefer a type that exists.

7.4 Why not a linear correction

The direct way to remove the composition from $\hat{\theta}_j(\tau)$ is a projection: regress the household effects on the observables $\mathbf{v}_i = (\mathbf{x}_i', \mathbf{z}_i')'$ with the percentile weights, and predict at the reference,

$$\hat{\theta}_j^L(\tau) = \hat{\theta}_j(\tau) + \hat{\pi}_j(\tau)'(\mathbf{v}^\circ - \bar{\mathbf{v}}(\tau)). \quad (35)$$

Table 7: Which object for which question

Object	Question	Typical uses
The group, $\theta_j(\tau)$	What a change of x_j for every household does to the τ -quantile	Universal policies; indicators defined at a quantile, such as a poverty line at a percentile; the incidence of a change across the distribution
The mean household, ref(mean)	Under het(z) , the average effect in the population	An aggregate benchmark; the size of the composition term at τ
A type that exists, the default	The effect for a household that can be observed; under het(qr) , whether it changes with the position	Results stated for a concrete household; heterogeneity along the distribution for a given household
Values given, ref(var=#)	The effect for a type chosen by the analyst	Targeting (the type a programme is aimed at); regional comparison with the rest of the type held fixed; a typology of households for a simulation of a marginal change

Under **het(z)** the household effects are affine in $\mathbf{v}_i$, the projection is exact, and (35) returns $b_j(\mathbf{x}^\circ, \mathbf{z}^\circ)$ at every τ . Under **het(qr)** the effects of the households at q_τ are $\beta_j(u_\tau(\mathbf{x}_i))$, which is not linear in $\mathbf{x}_i$, and the projection predicted at the reference keeps a bias that does not shrink with the sample. Two checks measure it.

On the Burkina Faso sample of Section 9, with urban as a regressor and the reference at the population means, the correction and the exact effect (34) agree in the interior of the distribution to within 0.02 (Table 8): the correction is a smoothed version of the exact curve, which jumps by the noise of each quantile regression. They part at the top, where the mean household found at $q_{0.9}$ is past its own 95th conditional percentile and the exact effect is held at $\hat{\beta}(0.95)$: the urban premium is 0.393 with the correction and 0.436 exact, the gap of crop farmers -1.149 and -1.241 . For household size the three curves agree to within 0.002.

In a design where the truth is known (Table 9), the bias of the correction is visible. For a type that exists, $\mathbf{x}^\circ = (5, 1)$, it is small at $n = 2,000$: measured in one sample of 200,000 households, it is $+0.0056$ for x_1 , a third of the standard error at $n = 2,000$, and it does not shrink with n while the standard error does; the exact estimator is within the noise of the truth. For the mean household the gap is larger, 1.2 standard errors at $n = 2,000$, and part of it is a matter of definition: with a binary regressor the mean household is not a type that exists, the projection averages the effects of the two types that do, and (34) evaluates the model at the fictitious one. The correction is 11 to 25 percent less variable than the exact estimator; that is what its bias buys. **gepwreg** uses (34).

Table 8: The effect of the mean household under `het(qr)`: the group (θ), the linear correction (L) and the exact effect (E), Burkina Faso 1998

τ	$\hat{u}_\tau^\circ$	Male head			Urban			Crop farmer		
		θ	L	E	θ	L	E	θ	L	E
0.1	0.064	0.002	0.014	0.015	0.252	0.242	0.243	-0.849	-0.825	-0.822
0.3	0.228	-0.027	-0.012	-0.004	0.276	0.262	0.255	-0.914	-0.885	-0.873
0.5	0.434	-0.054	-0.047	-0.037	0.305	0.299	0.284	-0.956	-0.944	-0.928
0.7	0.704	-0.076	-0.081	-0.090	0.329	0.337	0.332	-1.012	-1.031	-1.022
0.9	0.950	-0.086	-0.098	-0.104	0.361	0.393	0.436	-1.078	-1.149	-1.241

Regressors: household size, male head, urban, socio-economic group (crop farmer against public wage-earner); sampling weights; `het(qr)`, grid 0.05, 0.10, ..., 0.95. Reference: the population means of the regressors. L: (35); E: (34); at $\tau = 0.9$ the reference household is past the end of the grid and E is held at $\hat{\beta}(0.95)$. File `p11_profile.burkina.do`.

Table 9: The linear correction (L) against the exact estimator (E) where the truth is known, $\tau = 0.25$

	Type $\mathbf{x}^\circ = (5, 1)$			Mean household		
	truth	L	E	truth	L	E
<i>Estimate, one sample of $n = 200,000$</i>						
x_1	0.4488	0.4543	0.4474	0.4704	0.4898	0.4669
x_2	0.5732	0.5847	0.5745	0.6057	0.6389	0.6144
<i>Bootstrap standard error, $n = 2,000$</i>						
x_1		0.0174	0.0206		0.0155	0.0188
x_2		0.0803	0.0904		0.0800	0.1063

DGP: $y = \Phi^{-1}(u) + (0.4 + 0.4u)x_1 + (0.5 + 0.6u)x_2$, $u \sim U(0, 1)$ independent of (x_1, x_2) , $x_1 \sim U(1, 9)$, x_2 binary with $\Pr[x_2 = 1 | x_1] = \Lambda(-1 + 0.1x_1)$: the conditional quantile function is linear at every u . Truth: $\beta_j(u_\tau(\mathbf{x}^\circ))$ from 2,000,000 draws, with $u_\tau(\mathbf{x}^\circ) = 0.122$ for the type (5, 1) and 0.176 for the mean household. Bootstrap: pairs, $B = 400$, the whole procedure re-run with the bandwidth held at its sample value. File `p12_profile.dgp.do`.

7.5 Inference for the profile and for a given household

The delta method for a given household under `het(qr)`. Implicit differentiation of $\tilde{\mathbf{x}}^{\circ'}\boldsymbol{\beta}(u) = q_\tau$ on the segment of (34) gives the influence function

$$\psi_i(\hat{\theta}_j^\circ) = \psi_i(\beta_j^*) + \frac{\Delta_j}{D} \left[\psi_i(q_\tau) - \tilde{\mathbf{x}}^{\circ'}\psi_i(\boldsymbol{\beta}^*) - \sum_l \beta_l^* \psi_i(x_l^\circ) \right], \quad (36)$$

where $\boldsymbol{\beta}^* = (1-t)\boldsymbol{\beta}(u_g) + t\boldsymbol{\beta}(u_{g+1})$, $\Delta_j = \beta_j(u_{g+1}) - \beta_j(u_g)$, $D = \tilde{\mathbf{x}}^{\circ'}\{\boldsymbol{\beta}(u_{g+1}) - \boldsymbol{\beta}(u_g)\}$, $\psi_i(\boldsymbol{\beta}(u))$ is the influence function of the quantile regression at u , $\psi_i(q_\tau)$ that of the sample quantile, and the last sum runs over the reference values that are estimated means. Formula (36) is the right linearisation: in a design where every coefficient varies strongly with u ($n = 2,000$, $\tau = 0.25$, a bootstrap of the whole procedure), its standard error is within 4 to 13 percent of the bootstrap with the Powell kernel density of the quantile-regression sandwich, and within 1 to 11 percent with the Hendricks–Koenker density (Koenker, 2005), the largest gaps coming from draws in which $\hat{u}_\tau^\circ$ crosses a grid point, where the interpolated path changes slope. It rests on two estimates that are fragile in finite samples. The first is the sparsity of each quantile regression: on the Burkina Faso sample ($n = 8,478$, unweighted, $\tau = 0.25$, the default reference), the two standard densities miss the bootstrap in opposite directions and by up to 40 percent (household size: 0.90 of the bootstrap standard deviation with the Powell kernel, 0.59 with Hendricks–Koenker; file `p13_profile_delta.do`). The second is the local slope Δ_j , the difference of two adjacent quantile regressions, which is weakly identified: a steeper $\beta_j(u)$ also widens the conditional distribution of y , and with it the noise of every coefficient, so the t -statistic of a one-step difference grows with n and with the width of the step, not with the steepness. `gepwreg` reports (36) with a warning, and the bootstrap of the whole procedure (`boot()`) is the standard error to report.

The whole profile. Under `het(z)` the influence functions (27) at every quantile of the profile are computed on the same households, and their covariance across quantiles, $\hat{\Omega}_j$, is the joint covariance of the profile (with its Taylor linearisation under a design). Under `het(qr)` one loop of bootstrap draws serves every quantile: in each draw the whole procedure is re-estimated — the grid of quantile regressions, the conditional ranks, the reference household, and at each quantile its bandwidth and $\hat{q}_\tau$ — so the draws are draws of the curve, not of each point apart; at each quantile the standard error is exactly that of a separate estimation at that quantile with the same random numbers.

Simultaneous band. A pointwise interval at each of T quantiles covers the whole profile with a probability below its level. The simultaneous band is $\hat{\theta}_j(\tau) \pm \hat{c}\hat{\text{se}}(\tau)$, with $\hat{c}$ the quantile, at the chosen level, of $\max_\tau |\theta_j^*(\tau) - \hat{\theta}_j(\tau)|/\hat{\text{se}}(\tau)$: the sup- t band (Montiel Olea and Plagborg-Møller, 2019), of the kind Chernozhukov et al. (2013) give for function-valued effects. The distribution of the maximum comes from $\mathcal{N}(\mathbf{0}, \hat{\Omega}_j)$, by simulation, under `het(z)`, and from the bootstrap draws under `het(qr)`.

A test that the effect is constant across quantiles. Let $\mathbf{D} = \mathbf{A}\hat{\boldsymbol{\theta}}_j$ be the deviations of the profile from its mean over the T quantiles, $\mathbf{A} = \mathbf{I}_T - \mathbf{J}_T/T$, and $S = \max_t |D_t|/\text{sd}(D_t)$. Under the null that $\theta_j(\tau)$ is constant, $\mathbf{D}$ is centred, with covariance $\mathbf{A}\hat{\Omega}_j\mathbf{A}'$ under `het(z)`, simulated, and the covariance of the bootstrap deviations, centred at the estimate's, under `het(qr)`; the p -value is $(1 + \#\{S^* \geq S\})/(M + 1)$ over M draws, which moves in steps of $1/(B + 1)$ with B bootstrap draws. What the test says depends on the curve and on

the model. On the profile of the group under `het(z)` it is a test of composition: whether the households found at different quantiles differ enough, in the heterogeneity variables, to change the effect. On the curve of a given household under `het(qr)` it is a test that $\beta_j(u)$ is constant over the conditional ranks that household goes through, the location-shift hypothesis of [Koenker and Xiao \(2002\)](#) for one coefficient on a range of u .

8 Monte Carlo evidence

In Sections 8.1 to 8.3 the truth at τ is the estimand itself, $\theta_j(\tau) = \mathbb{E}[b_{ij} | y_i = q_\tau]$, computed exactly rather than simulated. Every DGP used there has a latent variable carrying the effect and a closed-form law of y given it, so $\theta_j(\tau) = \mathbb{E}_z[b_j(z)f(q_\tau|z)]/\mathbb{E}_z[f(q_\tau|z)]$ is a one-dimensional integral and q_τ solves $\mathbb{E}_z[F(q|z)] = \tau$ (`pwr_exact_truth.py`; under the survey design of Section 8.3 the cluster effects integrate out in closed form and the strata enter with the sampling weights). Table 3, which concerns the one-step estimator, keeps the reference used there: the mean effect of the households whose rank in y lies within ± 0.05 of τ . Standard deviations across seeds are in brackets.

8.1 The two-step estimator against RIF regression and the truth

Table 10 uses four DGPs with $n = 6,000$ and twenty-five seeds. In D3 and D1 the effect is $1 + r$ with $r \sim U(0, 1)$ observed, in H1 it is $1 + z$ with z observed and an error term, in U1 it is $1 + U$ with U unobserved and ordered with y , and an observed z that is irrelevant. The four differ in one thing only — where the heterogeneity of the effect comes from and whether the analyst can see it — and each estimator fails in the DGP built against it.

PWR-1, the one-step estimator, is between -0.009 and 0.73 for a truth between 1.10 and 1.90 , at every quantile of every DGP. Nothing else in the table is of that order, and the rest of this section is about differences in the third decimal.

Where the variable that generates the effect is observed and handed to PWR-2(z), it is not merely close, it is exact to the precision of the experiment: within 0.003 of the truth at all five quantiles of D3 and within 0.002 in D1, with a standard deviation across seeds of 0.003 to 0.007 and an influence-function standard error that matches it. In H1, where the outcome carries an error term, the bias is 0.006 to 0.011 in absolute value (negative at $\tau = 0.9$) — small, but $|t|$ between 1.8 and 3.3 over twenty-five seeds, so it is the smoothing bias of the second step and not noise.

D1 separates the estimators that need the conditional density of y given $\mathbf{x}$ from the one that does not. The level term $100r$ makes that density nearly flat, and both `het(qr)` and RIF regression become imprecise: standard deviations of 0.29 to 0.58 , against 0.003 to 0.005 for `het(z)`. Neither is biased there; they simply cannot measure anything.

U1 is the reverse experiment, and it is where the reading of this table has changed. The heterogeneity variable is unobserved, and the z handed to `het(z)` is irrelevant by construction: `het(z)` returns 1.503 at every quantile for a truth running from 1.18 to 1.80 , with t up to 60 . That is the price of a wrong observables model, stated as starkly as a design can state it. But `het(qr)`, which models the effect through the *conditional rank* rather than through observables, recovers the profile: $1.191, 1.300, 1.525, 1.683, 1.796$ against $1.183, 1.290, 1.533, 1.683, 1.800$, every cell within 0.011 and not one t above 1.6 in absolute value. When the source of the heterogeneity is unobserved but ordered with the outcome

— which is what the conditional-rank model assumes — **het**(**qr**) is doing exactly what it claims to do. RIF regression does not: at $\tau = 0.75$ it returns 2.005 for a truth of 1.683, a bias of 0.32 with $t = 27$, because the projection of $\mathbb{E}[\text{RIF} | \mathbf{x}]$ on $\mathbf{x}$ is linear and the true profile is not.

Which estimator wins, and where. No estimator dominates, and the table should not be read as if one did. Over the twenty cells the mean absolute bias is 0.054 for **het**(**z**), 0.023 for **het**(**qr**) and 0.046 for RIF-OLS, with mean standard deviations of 0.014, 0.125 and 0.159. Drop U1 and the ordering reverses for two of the three: **het**(**z**) falls to 0.003, while **het**(**qr**) *rises* to 0.029 and RIF-OLS to 0.034. The estimator that looks best on the average is the one carried by the single DGP built to defeat its competitor. That is the whole lesson of the table, and it is why the average is reported and then set aside.

The order in which to read the two models is in Section 4.4; what the table adds is where each one breaks. **het**(**qr**)’s weakness is H1, where the effect runs through a variable that is not a regressor and the linear conditional quantile model is an approximation ($t = 12.5$ at $\tau = 0.5$, $t = -9.6$ at $\tau = 0.9$). RIF-OLS runs neither model, which is its argument, and pays for it with a linearity that is a model too: besides the 0.32 of U1 it is 0.050 above the truth at $\tau = 0.25$ of D3 ($t = 5.5$) and 0.028 above at $\tau = 0.1$ ($t = 3.0$).

Table 10: The two-step estimator against RIF regression and the truth ($n = 6,000$; 25 seeds; mean [s.d.]

DGP	τ	Truth	PWR-1	PWR-2(z)	s.e. IF	PWR-2(qr)	RIF-OLS
D3: $y = 5r + (1 + r)x$	0.1	1.231	0.361 [0.023]	1.234 [0.006]	0.0061	1.243 [0.018]	1.259 [0.045]
	0.25	1.319	0.218 [0.059]	1.319 [0.007]	0.0067	1.325 [0.018]	1.369 [0.045]
	0.5	1.493	0.059 [0.013]	1.493 [0.005]	0.0056	1.495 [0.021]	1.492 [0.055]
	0.75	1.672	0.345 [0.060]	1.673 [0.006]	0.0060	1.673 [0.020]	1.676 [0.054]
	0.9	1.781	0.669 [0.045]	1.779 [0.005]	0.0052	1.777 [0.016]	1.754 [0.094]
H1: $y = 5 + (1 + z)x + z + e$	0.1	1.465	0.478 [0.039]	1.471 [0.013]	0.0146	1.456 [0.018]	1.458 [0.066]
	0.25	1.448	0.412 [0.082]	1.454 [0.017]	0.0149	1.476 [0.017]	1.459 [0.051]
	0.5	1.463	0.545 [0.309]	1.469 [0.016]	0.0146	1.501 [0.015]	1.494 [0.039]
	0.75	1.514	0.278 [0.030]	1.524 [0.016]	0.0142	1.529 [0.016]	1.525 [0.044]
	0.9	1.588	0.534 [0.040]	1.580 [0.021]	0.0148	1.554 [0.018]	1.579 [0.061]
D1: $y = 100r + (1 + r)x$	0.1	1.100	0.047 [0.106]	1.100 [0.004]	0.0034	1.180 [0.301]	1.171 [0.313]
	0.25	1.250	0.007 [0.102]	1.249 [0.005]	0.0052	1.351 [0.522]	1.356 [0.561]
	0.5	1.500	0.099 [0.239]	1.500 [0.005]	0.0052	1.546 [0.551]	1.525 [0.519]
	0.75	1.750	-0.009 [0.099]	1.750 [0.005]	0.0052	1.796 [0.510]	1.773 [0.575]
	0.9	1.900	0.177 [0.089]	1.899 [0.003]	0.0034	1.911 [0.294]	2.003 [0.369]
U1: $y = 5 + (1 + U)x + 3U$, z irrelevant	0.1	1.183	0.292 [0.027]	1.503 [0.027]	0.0263	1.191 [0.032]	1.189 [0.066]
	0.25	1.290	0.101 [0.019]	1.503 [0.027]	0.0263	1.300 [0.034]	1.326 [0.055]
	0.5	1.533	0.029 [0.011]	1.502 [0.027]	0.0263	1.525 [0.031]	1.527 [0.054]
	0.75	1.683	0.728 [0.175]	1.503 [0.027]	0.0263	1.683 [0.029]	2.005 [0.059]
	0.9	1.800	0.608 [0.034]	1.503 [0.027]	0.0263	1.796 [0.026]	1.839 [0.061]

$r, z, U \sim U(0, 1)$; $x \sim \mathcal{N}(0, 1)$ in D3, H1, D1 and $x \sim U(0, 2)$ in U1; $e \sim \mathcal{N}(0, 1)$. In D3, H1 and D1 the heterogeneity variable of **het**(**z**) is the one in the effect (r or z); in U1 it is an irrelevant $z \sim \mathcal{N}(0, 1)$. Truth: the exact $\theta(\tau)$ by quadrature. One-step: **rankdep**; two-step: the plug-in bandwidth (22); RIF-OLS: **rifhreg**. s.e. IF: mean of the influence-function standard error of **het**(**z**) over seeds. File **p4_twostep_vs_rif.do**.

8.2 The influence-function variance

Table 11 takes DGP H1 with $n = 4,000$ and 500 replications, for **het**(**z**) and for the profile along z (**rankvar**(**z**), whose object is $\mathbb{E}[1 + z | z = q_\tau^z] = 1 + \tau$). Two coverages are reported:

against the Monte Carlo mean of the estimator, which isolates the variance estimator from the smoothing term of (9), and against the estimator’s own object.

Consistency first. On a single sample of 400,000, `het(z)` returns 1.4644, 1.4622 and 1.5863 against the exact 1.4648, 1.4629 and 1.5881 — errors of 0.0004, 0.0007 and 0.0018. At that sample size the bandwidth has shrunk enough that the estimator should sit on $\theta(\tau)$, and it does.

For `rankvar(z)` everything is in order at every quantile: the ratio of the mean influence-function standard error to the Monte Carlo standard deviation is 1.00, the mean estimate is 1.104, 1.501 and 1.899 for 1.1, 1.5 and 1.9, and the coverage against $1 + \tau$ is 0.942–0.946.

For `het(z)` the ratio is 1.04 at $\tau = 0.1$ and 0.96 at $\tau = 0.5$, with coverages of 0.948 and 0.932 against the exact object. At $\tau = 0.9$ it is 0.835, and the coverage falls to 0.824. That failure comes with the plug-in — an earlier run of the same experiment with the Silverman bandwidth, which is not part of the replication, did not show it — and the reason is the one set out in Section 6.2: (29) conditions on h , and a bandwidth chosen on the sample adds a variance that a fixed- h influence function cannot contain. The size of that addition is measurable. Repeating H1 at $n = 4,000$ over 500 seeds, the standard deviation of $\hat{\theta}(\hat{h}^*)$ is 0.02023, while the standard deviation of $\hat{\theta}$ at the *same average width fixed in advance* is 0.01647: the act of choosing costs 23 percent in standard deviation at $\tau = 0.9$, 4 percent at $\tau = 0.5$ and nothing at $\tau = 0.1$. The implied ratio, $0.01647/0.02023 = 0.814$, accounts for the 0.835 observed here, and leaves nothing for an error in (29). The bootstrap, which re-chooses the bandwidth inside every resample, recovers it: over 150 seeds with $B = 200$ its mean standard error is 1.046, 0.994 and 1.030 times the Monte Carlo standard deviation at the three quantiles. File `pwr_bandwidth_variance.py`.

Table 11: Influence-function standard errors and coverage, DGP H1 ($n = 4,000$; 500 replications)

Estimator	τ	Object	Mean	MC s.d.	Mean s.e. IF	Ratio	Cov. (MC mean)	Cov. (object)
PWR-2(z)	0.1	1.4648	1.4697	0.0171	0.0178	1.038	0.946	0.948
	0.5	1.4629	1.4661	0.0186	0.0179	0.962	0.936	0.932
	0.9	1.5881	1.5732	0.0214	0.0179	0.835	0.890	0.824
PWR-cov(z)	0.1	1.1000	1.1044	0.0405	0.0405	1.001	0.942	0.942
	0.5	1.5000	1.5007	0.0214	0.0213	0.997	0.946	0.946
	0.9	1.9000	1.8986	0.0403	0.0405	1.004	0.946	0.942

Object: the exact $\theta(\tau) = \mathbb{E}[1 + z | y = q_\tau]$ for `het(z)`; $1 + \tau$ for `rankvar(z)`. Ratio: mean s.e. IF over MC s.d. Coverage of the nominal 95% interval against the Monte Carlo mean and against the object; Monte Carlo standard error on a coverage of 0.95: ± 0.010 . File `p5_coverage_mc.do`.

8.3 Complex survey design

Table 12 draws 300 samples from a superpopulation with 10 strata of 40 PSUs of 20 households ($n = 8,000$), a PSU effect in the level of y , an independent PSU effect in x , a stratum effect and sampling weights constant within strata, and estimates `het(z)` at $\tau = 0.25$ and 0.75 with the design declared. The Taylor standard error (31) is 1.09 and 1.04 times the Monte Carlo standard deviation, with a coverage of 0.973 and 0.943, slightly conservative (with 300 replications the Monte Carlo standard deviation itself has a relative standard error of four percent), which is also where the term of Section 6.2 goes: $\tau = 0.25$

and 0.75 are interior, the extra variance from choosing the bandwidth is a few percent, and the conservatism absorbs it. The influence-function standard error that ignores the design is 0.74 and 0.70 of the Monte Carlo standard deviation, with a coverage of 0.84 at both quantiles: intra-cluster correlation of the household effects is not removed by the kernel, and the design must be carried. The stratified cluster bootstrap on one sample ($B = 200$) gives 0.0191 and 0.0177 for a Monte Carlo standard deviation of 0.0182 and 0.0187. Against the exact $\theta(\tau)$ — here a one-dimensional integral over z mixed over the ten strata with the sampling weights, the cluster effects integrating out in closed form — the mean estimate is 0.004 and 0.006 high, a fifth and a third of a standard error, and a bias the window reference used previously could not have shown.

Table 12: Standard errors under stratified cluster sampling, DGP H1 with PSU and stratum effects ($n = 8,000$; 300 replications)

τ	Truth	Mean	MC s.d.	Taylor (PSU, strata)			IF ignoring the design			Cluster boot. one sample
				Mean s.e.	Ratio	Cov.	Mean s.e.	Ratio	Cov.	
0.25	1.4529	1.4570	0.0182	0.0198	1.087	0.973	0.0135	0.738	0.837	0.0191
0.75	1.5088	1.5150	0.0187	0.0194	1.037	0.943	0.0131	0.698	0.837	0.0177

$y = 5 + s_h + (1+z)x + z + a_c + e$, $x = c_c + \mathcal{N}(0, 1)$, $a_c, c_c \sim \mathcal{N}(0, 0.6)$ PSU effects, $s_h = 0.3(h - 5.5)/3$, weights $1 + 0.1h$. **het(z)**, design declared with **svyset**. Truth: the exact $\theta(\tau)$ for the weighted population; given z and the stratum, $y \sim \mathcal{N}(5 + s_h + z, 1.36[(1+z)^2 + 1])$. Cluster boot.: stratified cluster bootstrap on the first sample. Files `p6_svy_mc.do`, `pwr_exact_truth.py`.

9 Application: Burkina Faso 1998

9.1 Data and specification

The Burkina Faso 1998 household survey (INSID, 1998) is a two-stage stratified sample: 10 strata, 425 primary sampling units, 20 households per unit, $n = 8,478$ households with sampling weights. The outcome is the logarithm of expenditure per capita, with a weighted standard deviation of 0.84. The regressors are household size, a male-head indicator and the socio-economic group of the head (**gse**: public wage-earner, the reference; private wage-earner; artisan or trader; other earner; crop farmer; subsistence farmer, 57 percent of households; inactive). The heterogeneity variables of **het(z)** are the urban indicator (23 percent of households) and household size, so that the effect of every regressor may differ between urban and rural households and with household size, and the effect of household size may vary with size itself. All estimates carry the design:

```
svyset psu [pw=weight], strata(strata)
gepwreg lexp size male i.gse, per(0.25) het(urban size)
gepwreg lexp size male i.gse, per(0.25) het(qr) boot(50)
```

The first line gives the Taylor standard errors, the second the stratified cluster bootstrap. RIF regression (**rifhdreg** with the sampling weights) and the one-step estimator (**rankdep**) are reported for comparison.

9.2 The effect of household size and of a male head

Table 13 gives the effect of one more household member on log expenditure per capita at five quantiles. The one-step estimator is between -0.001 and -0.025 : at the median it is -0.0008 with a standard error of 0.0003 , “significant” and one hundredth of the effect. The two-step estimator on observables runs from -0.057 at the first decile to -0.109 at the ninth; the conditional-rank model gives -0.044 to -0.045 at every quantile; RIF regression runs from -0.032 to -0.062 . The profile of `het(urban size)` has a mechanical reading. Expenditure per capita is expenditure divided by size, so the marginal effect of a member on its logarithm contains $-1/\text{size}$ plus the economies of scale in total expenditure; the households at the first decile have 10.2 members on average and those at the ninth 4.6 (Table 14), so the effect must be smaller in absolute value at the bottom. The interacted model captures it through the quadratic term in size and reports it through the composition of the group; the conditional quantile model is linear in size at each conditional rank and returns the average slope. Here the observables model is the right one.

Table 13: Effect of household size and of a male head on log expenditure per capita, Burkina Faso 1998 (coefficient, standard error)

Estimator	$\tau = 0.1$	$\tau = 0.25$	$\tau = 0.5$	$\tau = 0.75$	$\tau = 0.9$
<i>Household size</i>					
PWR-2(z), <code>het(urban size)</code> , Taylor	-0.0566 (0.0020)	-0.0613 (0.0023)	-0.0742 (0.0029)	-0.0908 (0.0038)	-0.1087 (0.0048)
PWR-2(qr), cluster bootstrap	-0.0446 (0.0030)	-0.0435 (0.0024)	-0.0441 (0.0020)	-0.0451 (0.0020)	-0.0454 (0.0021)
RIF-OLS (<code>rifhdreg</code>)	-0.0320 (0.0028)	-0.0390 (0.0020)	-0.0465 (0.0019)	-0.0561 (0.0025)	-0.0619 (0.0041)
PWR-1, rank of y (<code>rankdep</code>)	-0.0025 (0.0010)	-0.0031 (0.0006)	-0.0008 (0.0003)	-0.0043 (0.0007)	-0.0249 (0.0034)
<i>Male head</i>					
PWR-2(z), <code>het(urban size)</code> , Taylor	0.2276 (0.0879)	0.2097 (0.0798)	0.1614 (0.0590)	0.1116 (0.0414)	0.0692 (0.0349)
PWR-2(qr), cluster bootstrap	-0.0050 (0.0382)	-0.0292 (0.0378)	-0.0728 (0.0331)	-0.0814 (0.0357)	-0.0989 (0.0439)
RIF-OLS (<code>rifhdreg</code>)	-0.0124 (0.0333)	-0.0238 (0.0289)	-0.0989 (0.0332)	-0.1441 (0.0571)	0.0049 (0.0956)
PWR-1, rank of y (<code>rankdep</code>)	0.0104 (0.0210)	-0.0021 (0.0122)	-0.0102 (0.0060)	-0.0060 (0.0094)	0.0049 (0.0275)
n_{eff} (two-step)	1,866	2,095	1,979	2,601	1,939

Regressors: size, male head, socio-economic group. Taylor: linearisation with PSU and strata. Cluster bootstrap: $B = 50$, PSUs within strata. RIF-OLS: `rifhdreg` with sampling weights, robust s.e. One-step: Taylor s.e. File `p7_burkina.do`.

The male-head effect, the lower panel of Table 13, is the case where the two heterogeneity models disagree. The observables model gives $+0.23$ at the first decile falling to $+0.07$ at the ninth; the conditional-rank model gives -0.01 falling to -0.10 ; RIF regression -0.01 to -0.14 up to the third quartile and 0.00 at the ninth decile, with a standard error of 0.10 there. Table 14 shows where the observables profile comes from: the reference effect of a

male head is -0.008 , and the contribution of the interaction with size is 0.24 at the first decile and 0.11 at the ninth, a coefficient of 0.0235 per member times a mean size of 10.2 and 4.6 . The whole male premium of the observables model is a linear interaction with household size, estimated on the full sample. The profile along size of Table 15, which is local in size and needs no interaction, does not support a premium that rises with size: it is 0.24 for the smallest households ($\tau_{\text{size}} = 0.1$, two members), 0.03 – 0.09 in the middle and -0.05 for the largest, with standard errors of 0.05 – 0.13 . The linear interaction is a poor description of how the male-head effect varies with size, and the observables model extrapolates it to the large households at the bottom of the distribution. Here it is the conditional-rank model and RIF regression that agree, and the disagreement is what a practitioner should look for: when $\text{het}(\mathbf{z})$ and $\text{het}(\mathbf{qr})$ differ, the heterogeneity model, not the weights, is in question, and the profile along the suspected variable (**rankvar**) is the check that costs nothing.

Table 14: Composition of the group at τ and contribution of household size to the effects, **het(urban size)**

τ	Urban share		Household size		Contribution of size to	
	at τ	population	at τ	population	effect of size	effect of male
0.1	0.060	0.226	10.16	7.58	0.1445	0.2390
0.25	0.083	0.226	9.46	7.58	0.1397	0.2225
0.5	0.142	0.226	7.55	7.58	0.1265	0.1776
0.75	0.338	0.226	5.92	7.58	0.1090	0.1391
0.9	0.555	0.226	4.65	7.58	0.0900	0.1093

From **e(decomp)**. Contribution of size to the effect of x_j : $\hat{d}_{j,\text{size}} \overline{\text{size}}(\tau)$, plus the own terms of (13) for $x_j = \text{size}$. File **p7_burkina.do**.

9.3 The socio-economic groups

The effect of the head’s group relative to a public wage-earner is the place where the two representations of the same object part company most visibly. The two-step estimators agree on the ordering and on the scale of the gaps: -0.7 to -0.9 (**het(urban size)**) and -1.0 to -1.3 (**het(qr)**) for crop farmers, -1.0 to -1.2 and -1.2 to -1.6 for subsistence farmers, larger in absolute value at the top of the distribution. RIF regression gives -0.16 and -0.33 for the two groups at the first decile and -3.3 and -3.6 at the ninth. The RIF coefficient of an indicator is the difference in the probability of being above q_τ between the two groups divided by the density $f(q_\tau)$, and in the upper tail the density is small: -3.3 is not a gap in log expenditure but the effect on the ninth decile of moving a marginal share of households from one group to the other, scaled by a small density. The two representations of the UQPE agree on the object and not on the number when the RIF projection is far from linear, and the two-step estimator, which reports the average of household-level effects, stays on the scale of the outcome.

9.4 The profile along household size

Table 15 is **rankvar(size)**: the effect of the other regressors among households at the τ -quantile of size, from two members ($\tau = 0.1$) to fourteen ($\tau = 0.9$). The urban premium is between 0.24 and 0.43 at all sizes with no trend, which is consistent with the small

size interaction of the urban effect in the observables model; the gaps of the farming groups narrow with size (-1.32 to -0.91 for subsistence farmers and -1.04 to -0.67 for crop farmers, from the smallest to the largest households); the male-head effect has been discussed. The standard errors are the influence-function ones with the design; they are larger than at the quantiles of y because the groups are smaller (n_{eff} from 905 to 2,522, the ties in size making the kernel groups uneven).

Table 15: Profile of the effects along household size, `rankvar(size)` (coefficient, s.e.)

Variable	$\tau_{\text{size}} = 0.1$	0.25	0.5	0.75	0.9
Male head	0.239 (0.057)	0.085 (0.050)	0.026 (0.073)	0.049 (0.095)	-0.055 (0.126)
Urban	0.374 (0.101)	0.244 (0.059)	0.434 (0.063)	0.259 (0.062)	0.362 (0.057)
Crop farmer	-1.043 (0.189)	-0.815 (0.100)	-0.849 (0.096)	-0.864 (0.110)	-0.670 (0.097)
Subsistence farmer	-1.319 (0.110)	-1.050 (0.077)	-1.003 (0.077)	-1.066 (0.101)	-0.911 (0.087)
Inactive	-0.908 (0.109)	-0.830 (0.106)	-0.924 (0.119)	-0.768 (0.127)	-0.714 (0.096)
n_{eff}	905	1,738	1,409	1,201	2,522

Regressors: male, urban, socio-economic group; weights on the rank of household size; Taylor s.e. File `p7_burkina.do`.

9.5 The urban premium at the initial quantile

Table 16 estimates the urban premium with urban as a regressor and two heterogeneity models, at the quantile of observed log expenditure and at the quantile of y_0 , log expenditure net of the urban contribution $\hat{b}_{i,\text{urban}} \text{urban}_i$ (reference value zero: the outcome of the household as if it were rural). With `het(size)` the premium is 0.35 at the bottom and 0.32 at the top; with `het(qr)` it is 0.25 at the bottom and 0.37 at the top: the two models give different profiles for the same average, and the conditional-rank model, which lets the premium vary with the household's position in the conditional distribution, finds it larger for the households that are rich given their characteristics (a rise that the test of Section 7.5 does not distinguish from a constant with $B = 50$; Section 9.6). For both models the effect at the initial quantile is within 0.007 of the effect at the observed quantile, and the mean size of the group moves by at most 0.13 members. The re-ranking on y_0 is weak here because the urban contribution, with a standard deviation of 0.15 (0.35 times the standard deviation of an indicator with mean 0.23), is small relative to the dispersion of log expenditure (0.84); the objects of Definitions 1 and 2 differ when the covariate accounts for a large share of the variance of the outcome, as in the Monte Carlo of Section 8 (an effect $1+z$ on $x \sim \mathcal{N}(0, 1)$, where they differ by 0.06 at the first decile) or for a programme with a large effect on the outcome.

9.6 The profile across quantiles: the group and a given household

Figures 1 and 2 draw the profile of four effects at 37 quantiles from 0.05 to 0.95, with urban as a regressor (regressors: household size, male head, urban, socio-economic group), under the two heterogeneity models; they are examples 8 and 9 of the help of `gepwreg`.

Under `het(urban size)` (Figure 1) the reference is the mean household of the population, so its flat curve is also the average effect: -0.078 (s.e. 0.003) for one more member, against -0.061 at the first quartile, where the households are larger and more often rural (9.5 members and 8 percent urban, against 7.6 and 23 percent in the population). The profile

Table 16: The urban premium at the initial quantile and at the quantile of log expenditure

Heterogeneity model	τ	At the initial quantile	s.e. (boot.)	At the quantile of y	Size at τ of y_0	Size at τ of y
PWR-2(z), het(size)	0.1	0.3524	0.0418	0.3527	10.14	10.18
	0.25	0.3471	0.0407	0.3480	9.33	9.46
	0.5	0.3363	0.0402	0.3358	7.68	7.61
	0.75	0.3253	0.0422	0.3247	6.01	5.92
	0.9	0.3160	0.0454	0.3163	4.60	4.64
PWR-2(qr)	0.1	0.2514	0.0420	0.2520	.	.
	0.25	0.2674	0.0389	0.2694	.	.
	0.5	0.3026	0.0433	0.3052	.	.
	0.75	0.3366	0.0512	0.3363	.	.
	0.9	0.3669	0.0578	0.3605	.	.

Regressors: urban, size, male, socio-economic group. **initial(urban)**, reference value zero (the default). s.e.: cluster bootstrap, PSUs within strata, $B = 200$ (**het(size)**) and $B = 50$ (**het(qr)**). Size at τ : mean household size of the group, from **e(zbar_tau)**. File **p7_burkina.do**.

of the size effect runs from -0.06 at $\tau = 0.05$ to -0.115 at $\tau = 0.95$, and the test rejects a constant effect ($p = 0.000$), as it does for crop farmers; it does not for the male head ($p = 0.118$) or the urban premium ($p = 0.592$). Under this model these rejections are statements about composition: the size of the households and their location change along the distribution, and the model makes the effect depend on both.

Under **het(qr)** (Figure 2) the reference is a type that exists — household size at its mean, a male head, rural, a subsistence farmer — found at the first quartile at its conditional rank 0.298, where its effects are within 0.02 of those of the group. The tests of a constant effect give $p = 0.67$ for the group and 0.77 for the given household for household size, 0.20 and 0.26 for the male head, and 0.16 and 0.26 for the urban premium: no evidence that these effects change with the position of a household. The gap of crop farmers widens along the distribution, from -0.84 to -1.09 for the group, with $p = 0.039$, and 0.059 for the given household: weak evidence at most, with $B = 50$ draws (p in steps of 0.02). From $\tau = 0.825$ the reference household would need a conditional rank above 0.95 to be found at q_τ , and its curve is held at the last quantile of the grid: the flat end of the dashed curves is the grid, not a finding.

The two figures do not contradict each other on household size, where the first finds a profile and the second none. **het(qr)** lets the effect vary with the conditional rank and finds no evidence that it does; **het(urban size)** does not let it vary with the rank, and lets it vary with the size and the location of the household, which change along the distribution. The variation of the group's effect in the first figure is the composition in size discussed with Table 13, which a model linear in size at every conditional rank cannot represent. The urban premium of **het(qr)**, which rises from 0.25 at the bottom to 0.37 at the top as in Table 16, is not distinguished from a constant across the 37 quantiles ($p = 0.16$ for the group, with $B = 50$).

10 Conclusion

The percentile weights select the households at a quantile of the outcome correctly, and a regression run with them measures nothing about the effect, because the outcome does not vary inside the group it selects. That was the estimator of the earlier papers, and no bandwidth, correction or reweighting removes the attenuation.

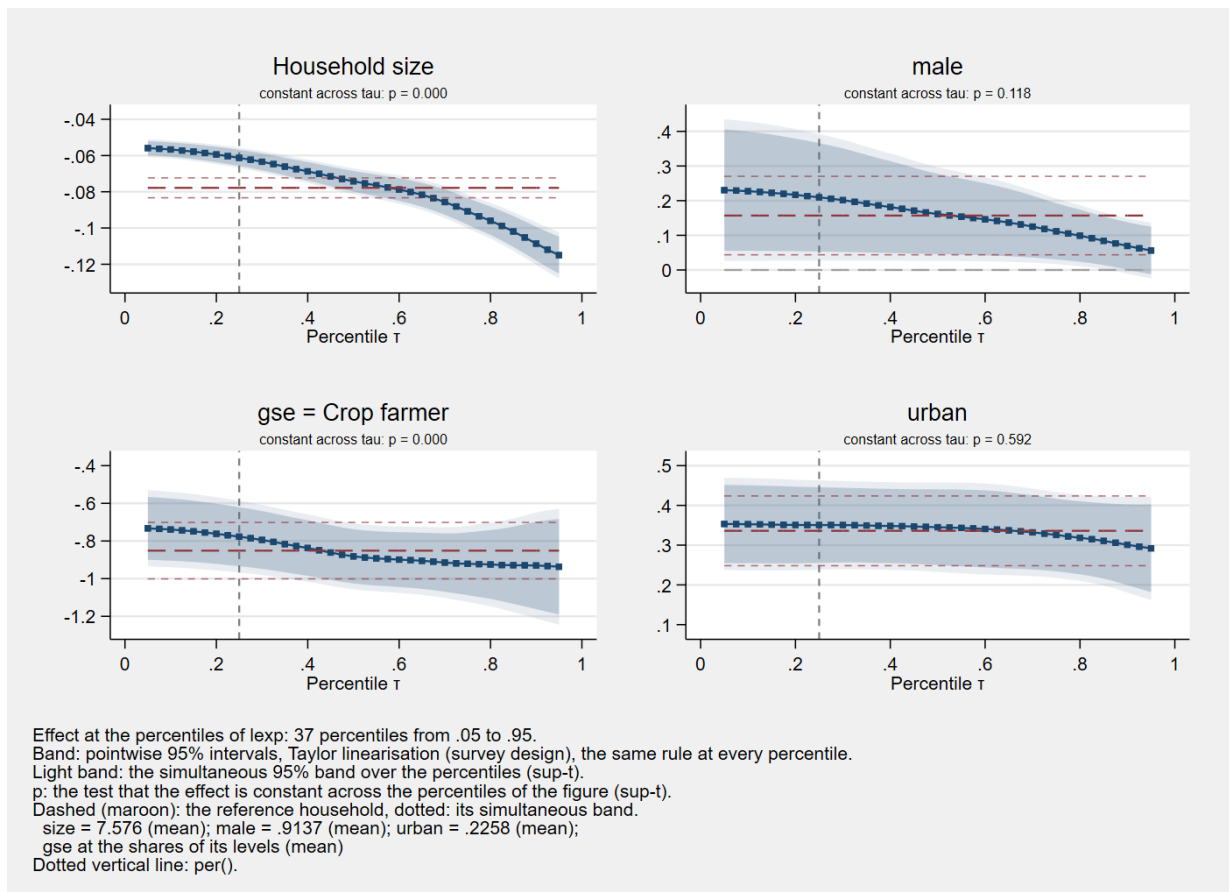


Figure 1: The profile of four effects across quantiles under `het(urban size)`, and the mean household of the population

Dark band: pointwise 95 percent intervals; light band: the simultaneous 95 percent band; dashed: the mean household of the population, which under this model is the average effect, with its simultaneous band dotted; p under each title: the test that the effect is constant across the 37 quantiles. Taylor linearisation with PSU and strata. Example 8 of the help of `gepwreg`: `gepwreg lexp size male urban i.gse, per(0.25) het(urban size) ref(mean) graph(size male 5.gse urban, ref uniform)`. File `p14.profile_figures.do`.

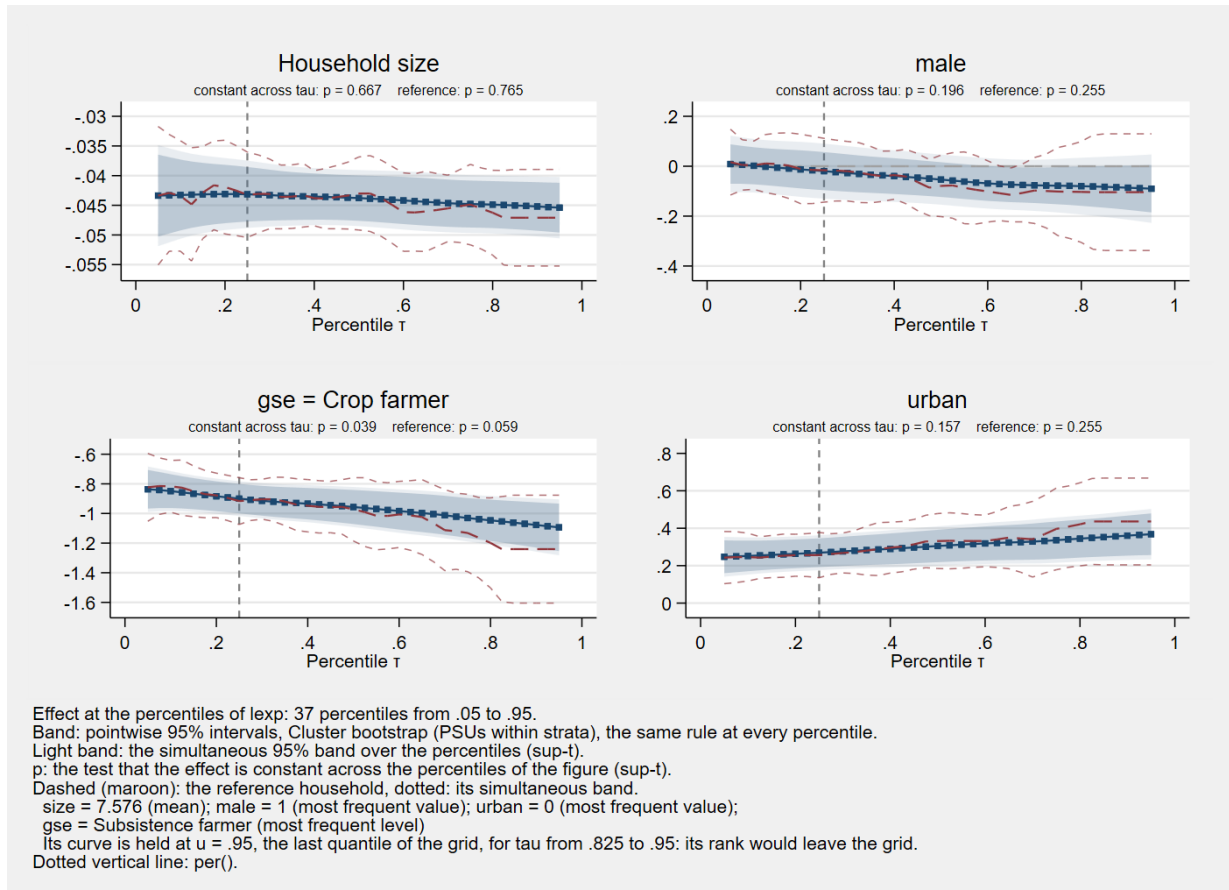


Figure 2: The profile of four effects across quantiles under `het(qr)`, and a given household

As Figure 1. Reference household: size at its mean, male head, rural, subsistence farmer; p : the test for the group, then for the reference household. Cluster bootstrap, PSUs within strata, $B = 50$, one loop of draws for every quantile. Example 9 of the help of `gepwreg`: `gepwreg lexp size male urban i.gse, per(0.25) ref(default) boot(50) graph(size male 5.gse urban, ref uniform)`. File `p14.profile_figures.do`.

The estimator that replaces it keeps the weights and changes what they are applied to. Household-level effects from a heterogeneity model, averaged with the percentile weights, converge to the effect at the quantile, which is the unconditional quantile partial effect. The conditional-rank model reaches heterogeneity that observables cannot; RIF regression, which needs no model of the effect, is the check on both — a check that is itself a linear projection, and fails where the profile curves. The Monte Carlo puts the two-step estimator at the truth with standard errors at nominal coverage, and the application shows both the gain (the size effect, with its composition explained) and the price (a male-head effect that depends on the heterogeneity model, and the diagnostic that reveals it).

The profile of the effect across quantiles mixes the composition of the group with the heterogeneity of the effect, because the group is defined by the outcome. Following a given household separates the two: under the observables model the whole profile is composition, by construction; under the conditional-rank model the household’s curve is the matched coefficient at its own conditional rank, estimated exactly rather than by a linear correction, which is inconsistent there. The profile comes with a simultaneous band and a test of constancy, and the application shows why both are needed: a profile of household size that is composition, and an urban premium that rises along the distribution without being distinguishable from a constant.

Four things are left open. A locally linear smoother in rank space, which would remove the first-order bias the kernel carries at the extreme quantiles, where the plug-in bandwidth is also least reliable. The influence function of the conditional-rank model and of the initial-quantile object, for which the bootstrap is used here, the first being available in [Alejo et al. \(2025\)](#) for their version of the estimator; for a given household under that model the delta method is derived, and what it lacks is an estimate of the sparsity reliable in samples of survey size. A third heterogeneity model, in which the joint distribution of the household coefficients is identified from the conditional moments of the outcome given the covariates in the manner of [Hildreth and Houck \(1968\)](#) and [Beran and Hall \(1992\)](#), and each household’s effect is the posterior mean of its own coefficient: it needs two least squares fits and no quantile regression, it recovers the effect where the observables model is flat, and it fails — qualitatively, not only in level — as soon as the effect depends on the level of a regressor, which is the case in the application above. And finite changes of a covariate with re-ranking, for which the household-level effects and the initial-quantile machinery are in place.

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A Proofs

Equation (2). Let $F_t(q) = \Pr[y + tb \leq q]$. Conditioning on b , $F_t(q) = \mathbb{E}[F_{y|b}(q - tb | b)]$, so $\partial F_t(q)/\partial t|_{t=0} = -\mathbb{E}[b f_{y|b}(q | b)] = -f(q) \mathbb{E}[b | y = q]$ by Bayes’ rule, and $\partial F_0(q)/\partial q = f(q)$. Differentiating $F_t(Q_\tau(t)) = \tau$ at $t = 0$ gives $dQ_\tau/dt = -(\partial_t F)/(\partial_q F) = \mathbb{E}[b | y = q_\tau]$. $\square$

Proof of Proposition 1. Write $p = F(y)$, $\tilde{y}(p) = F^{-1}(p)$ and $\tilde{m}(p) = m(F^{-1}(p))$, and let $\mathbb{E}_w[\cdot]$ denote expectation under the density proportional to $K((p - \tau)/h)$ on $[0, 1]$. Conditional on p , $y = \tilde{y}(p)$ is fixed and $\mathbb{E}[x | p] = \tilde{m}(p)$, so $\text{Cov}_w(x, y) = \mathbb{E}_w[(x - \mathbb{E}_w x)(\tilde{y} - \mathbb{E}_w \tilde{y})] = \mathbb{E}_w[(\tilde{m} - \mathbb{E}_w \tilde{m})(\tilde{y} - \mathbb{E}_w \tilde{y})] = \text{Cov}_w(\tilde{m}, \tilde{y})$. For $h \rightarrow 0$ the weight is concentrated on $|p - \tau| = O(h)$; expanding $\tilde{y}(p) = q_\tau + (p - \tau)/f(q_\tau) + O((p - \tau)^2)$ and $\tilde{m}(p) = \tilde{m}(\tau) + \tilde{m}'(\tau)(p - \tau) + O((p - \tau)^2)$ with $\tilde{m}'(\tau) = m'(q_\tau)/f(q_\tau)$, and using that the odd central moments of a symmetric kernel vanish, $\text{Cov}_w(\tilde{m}, \tilde{y}) = \tilde{m}'(\tau)\text{Var}_w(p)/f(q_\tau) + O(h^4)$ with $\text{Var}_w(p) = \sigma_K^2 h^2 + O(h^4)$. Hence $\text{Cov}_w(x, y) = \sigma_K^2 h^2 m'(q_\tau)/f(q_\tau)^2 + O(h^4)$. By the law of total variance under the weighting, $\text{Var}_w(x) = \mathbb{E}_w[v(\tilde{y}(p))] + \text{Var}_w(\tilde{m}(p)) = v(q_\tau) + O(h^2)$. Dividing gives (6). The bound $|\text{Cov}_w(x, y)| \leq \text{sd}_w(x) \text{sd}_w(y)$ is Cauchy–Schwarz, and $\text{sd}_w(y) = \text{sd}_w(\tilde{y}(p)) = O(h)$ because $\tilde{y}$ is Lipschitz at q_τ with constant $1/f(q_\tau)$. With a vector of regressors, $\text{Cov}_w(\mathbf{x}, y)$ is the vector $\sigma_K^2 h^2 \mathbf{m}'(q_\tau)/f(q_\tau)^2 + O(h^4)$ with $\mathbf{m}(y) = \mathbb{E}[\mathbf{x} | y]$, $\text{Var}_w(\mathbf{x}) \rightarrow \mathbf{V}(q_\tau) = \text{Var}(\mathbf{x} | y = q_\tau)$, and $\mathbf{b}_w = \mathbf{V}(q_\tau)^{-1} \mathbf{m}'(q_\tau) \sigma_K^2 h^2 / f(q_\tau)^2 + O(h^4)$. $\square$

Proof of Proposition 2. Under $\mathbb{E}[x | y] = a + \gamma y$ and $\text{Var}(x | y) = v$, any weighting that depends on y alone leaves the conditional distribution of x given y unchanged, so $\text{Cov}_w(x, y) = \gamma \text{Var}_w(y)$ and $\text{Var}_w(x) = \gamma^2 \text{Var}_w(y) + v$. Hence $b_w = \gamma \text{Var}_w(y) / \text{Var}_w(x)$, and with $w \equiv 1$, $\beta = \gamma \text{Var}(y) / \text{Var}(x)$. Taking the ratio gives (7). $\square$

Proof of Proposition 3. (i) As $n \rightarrow \infty$, $\hat{p}_i \rightarrow p_i$ uniformly and the weighted mean converges to $\mathbb{E}_w[b] = \int K((p - \tau)/h) g(p) dp / \int K((p - \tau)/h) dp$. Expanding g to second order at τ and using the symmetry of K , $\mathbb{E}_w[b] = g(\tau) + \frac{1}{2} g''(\tau) \text{Var}_w(p) + O(h^4)$ with $\text{Var}_w(p) = \sigma_K^2 h^2 + O(h^4)$, which is (9); and $g(\tau) = \mathbb{E}[b | F(y) = \tau] = \mathbb{E}[b | y = q_\tau]$. (ii) $\mathbb{E}[w_i | \mathbf{x}_i = \mathbf{x}] = \int K((F(y) - \tau)/h) f(y | \mathbf{x}) dy = \int K((p - \tau)/h) \frac{f(F^{-1}(p) | \mathbf{x})}{f(F^{-1}(p))} dp$ after the change of variable $p = F(y)$. As $h \rightarrow 0$ this is $h \int K \cdot f(q_\tau | \mathbf{x}) / f(q_\tau) + o(h)$, while $\mathbb{E}[w_i] = h \int K + o(h)$; the ratio converges to $\omega_\tau(\mathbf{x})$. $\square$

The rank term of (28). Write $\mu_l = \sum_i f_i w_i(\hat{p}_i) z_{il} / \sum_i f_i w_i(\hat{p}_i)$ as a functional of the empirical distribution. Adding mass ε at household j has two effects. The direct one adds household j 's own term to the numerator and denominator, contributing $n f_j w_j(z_{jl} - \mu_l) / S$ to the influence function. The indirect one shifts the rank of every household i with $y_i \geq y_j$ by εf_j (normalised weights), hence its weight by $\varepsilon f_j \dot{w}_i$, which changes μ_l by $\varepsilon f_j \sum_i \dot{w}_i f_i(z_{il} - \mu_l) \mathbf{1}[y_i \geq y_j] / S = \varepsilon f_j \sum_i a_{il} \mathbf{1}[y_i \geq y_j] / S$. Centring over j ($\mathbb{E}_j \mathbf{1}[y_i \geq y_j] = p_i$) gives (28). After sorting on y , $\sum_i a_{il} \mathbf{1}[y_i \geq y_j]$ for all j is one reverse cumulative sum. $\square$

B Replication

The replication material for this paper — the programmes listed below, the Burkina Faso extract they use, and the script that rebuilds the tables — is in the `replication` folder of the repository that accompanies it, <https://github.com/aabbdd12/gepwreg/tree/v15/replication>, and is archived with this paper on Zenodo. The same branch, <https://github.com/aabbdd12/gepwreg/tree/v15>, holds version 1.5.0 of `gepwreg`; the README of the replication folder gives the two lines that install it and bring the survey extract. The programmes `p1` to `p10` and the checks are those of the previous version of this paper (branch `v14` of the same repository) and were run with version 1.4, which their headers

name; version 1.5.0 changes no result of version 1.4 except for a binary regressor that is also a heterogeneity variable, which none of them uses. The programmes `p11` to `p14` produce the results of Sections 7 and 9.6 and need version 1.5.0, except `p13`, which programs its estimator with official quantile regressions and does not call `gepwreg`. Each programme that calls `gepwreg` begins with `which gepwreg`, which reports the version it found. Every table but Tables 5 and 6 to 9 is generated by `make_tables.py` from the CSV files the programmes write; Table 5 and the numbers of Section 8.2 on the choice of the bandwidth are the output of `pwr_bandwidth_check.py` and `pwr_bandwidth_variance.py`, recorded in the header of each file; Tables 8 and 9 are printed by `p11` and `p12`, and Tables 6 and 7 contain no estimate. The three programmes that compare against RIF regression need `rifhdreg`, which is part of the `rif` module (Rios-Avila, 2018) and installs with `ssc install rif`; the version used here is 2.55 of August 2021, and the copy archived with the Stata Journal article is an earlier one.

File	Tables
<code>p1.onestep_attenuation.do</code>	3, 2
<code>p2.closed_doors.do</code>	the three repairs of Section 3.5
<code>p3.replications.do</code>	the examples of the 2016 and 2023 notes, Section 3.4
<code>p4.twostep_vs_rif.do</code>	10
<code>p5.coverage_mc.do</code>	11
<code>p6.svy_mc.do</code>	12
<code>p7.burkina.do</code>	13–16
<code>p7d.tableD.do</code>	16 alone, without rerunning <code>p7.burkina.do</code>
<code>p8.merr.do</code>	Section 5.3
<code>p9.merr_curvature.do</code>	Section 5.5
<code>p10.rif_gap.py</code>	4
<code>p11.profile_burkina.do</code>	8
<code>p12.profile_dgp.do</code>	9 (with <code>p12.core_mata.do</code>)
<code>p13.profile_delta.do</code>	the delta-method standard errors of Section 7.5 (with <code>p13a.E.draws.do</code> and <code>p13b.E.densities.do</code>)
<code>p14.profile_figures.do</code>	Figures 1 and 2, the numbers of Section 9.6
<code>p8.truth.py</code>	the values of $\theta(\tau)$ of Sections 5.3 and 5.5
<code>pwr_bandwidth_check.py</code>	5
<code>pwr_bandwidth_variance.py</code>	the variance of the bandwidth choice, Section 6.2
<code>pwr_exact_truth.py</code>	the exact $\theta(\tau)$ of Section 8
<code>gepwreg_gmm_check.do</code>	Section 4.7
<code>gepwreg_test15.do</code>	tests of the command against closed forms and known effects
<code>make_tables.py</code>	the tables of the programmes <code>p1</code> to <code>p10</code> , from the CSV files they write